EXHIBIT E

United States v. CEMEX, Inc.
Civil Action No. 09-cv-00019-MSK-MEH
U.S. District Court for the District of Colorado

Expert Report of Gerald L. Young

May 10, 2011

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ON BEHALF OF CEMEX, INC.

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1.0 BACKGROUND

1.1 Information Required by the Federal Rules of Civil Procedure

The following is a list of items required by the Federal Rules of Civil Procedure.

1. This report contains my opinions, conclusions and the reasons therefore, including on-site observations.

2. Appendix A contains a statement of my qualifications and the compensation received for working on this matter.

3. Appendix B lists the references, data or other information considered in forming my opinions.

4. Appendix C lists publications I have authored in the past ten years.

5. I have never before provided testimony as an expert at a trial or deposition.

The opinions expressed in the report are my own and are based on the data and facts available to me at the time of writing. Should additional relevant or pertinent information become available, I reserve the right to supplement the discussion and findings in my report.

1.2 Purpose of this Report

I have been requested by attorneys at Crowell & Moring LLP, representing CEMEX, Inc., to provide information and opinions on the following subjects:

1. General information about Portland cement, including a description of the manufacturing processes in general and the manufacturing processes at the CEMEX Lyons, Colorado cement plant in particular, including related atmospheric emissions.

2. A description of the general components of Portland cement plants and how they operate and, in particular, how the Lyons plant operates.

3. A description of the purpose, nature, and anticipated effects of changes made by CEMEX to the Lyons plant kiln system between 1997 and 2000.

In formulating this report, I have relied on forty-one years of experience. Twenty years of this experience in the Portland cement industry was spent working as chief chemist, production engineer, superintendent of process and production, and manager of energy and environmental affairs for major cement producers. I was also responsible for a research project in the mid-1980s to determine what factors influenced the formation of

NOx in a long, dry process cement kiln. My experience also includes five years with a company that had a patented wet sulfur dioxide scrubber that used cement kiln dust as the scrubbing reagent. Most recently, I have spent sixteen years as a process and environmental consultant to the cement industry. I have also authored or co-authored more than 25 papers that cover cement manufacturing and emissions control from cement manufacturing facilities.

I had access to numerous documents produced by the United States and by CEMEX in this case, including all of the documents cited by the experts of the United States. I have reviewed these documents and noted those that appear to be relevant to developing my opinions. The documents I have considered are listed in Appendix B to this report.

I toured the Lyons plant accompanied by professional staff from CEMEX and by an attorney from Crowell & Moring on April 14, 2011. Also, approximately ten years ago, I worked on a small project at the Lyons facility to determine the feasibility of using waste heat from the cement kiln in the raw material dryer to reduce the amount of fossil fuels used in the dryer.

1.3 Summary of Opinions

The following is a summary of the author's opinions regarding changes and efficiency improvements made by CEMEX (Southdown) during the period of 1997 through 2000.

1. During 1997, CEMEX made two efficiency improvements to their one-stage preheater, calciner pyroprocessing system. One improvement was made to the calciner internal geometry in order to reduce "burnover." The improvements to the calciner internal geometry were made in the spring of 1997 and reduced the cross sectional area of the calciner to increase turbulence for better air-fuel mixing. Reducing the cross section and increasing turbulence would cause the static pressure drop across the calciner to increase. At a given ID fan power consumption (amp load), this would reduce the volume of air that the fan can pull through the system and accordingly reduce the total amount of fuel that could be consumed in the pyroprocess. This would cause an overall decrease in fuel NOx emissions as well as a decrease in thermal NOx emissions in the kiln. Moreover, to the extent CEMEX was able to shift its fuel split in favor of the calciner after this improvement, any such shift would have further decreased thermal NOx formation in the kiln. Improved calcination in the calciner would also have lowered the thermal load on the kiln and further reduced thermal NOx.

   CEMEX also began using oxygen enrichment (OE) in the calciner to improve combustion conditions and to increase clinker production. CEMEX conducted tests in 1997 and 1998 to determine the effects of OE on emissions. Both tests indicated that NOx actually decreased. Accordingly, CEMEX was reasonable in concluding that it was not required to do a PSD review.

The actual NOx emissions in 1997 and 1998 were both lower than the NOx emissions in 1996 while the clinker production rates were higher. These facts further substantiate the reasonableness of CEMEX's belief that NOx emissions would be reduced with the efficiency improvement to the calciner and the use of OE. In summary, these efficiency improvements allowed CEMEX to increase clinker production without increasing NOx emissions on an annual mass basis.

2. In 1997, the 1250 hp ID fan motor suffered a catastrophic failure. The pyroprocessing system was shut down because the ID fan is an integral part of the system. This constituted an emergency repair, not a planned project to increase clinker production.

   Any increase in rpm and air flow that was realized at the location of the fan because of the larger motor would have resulted in a significantly smaller increase in air flow through the pyroprocessing system due to the fact that the air flow rate at the fan is not fully realized at the fuel firing locations. For example, ambient air leaks and water sprays in the conditioning tower will make up a part of any increase in air flow through the fan. Also, any additional fuel that could have been used would have been split approximately 50% to the calciner. The effect on total NOx emissions would have been minimal.

   Based on the author's experience, plant personnel would have been reasonable not to have considered the emergency replacement of the ID fan motor to be a project to increase production or a PSD triggering modification.

3. Between 1997–1999, minor improvements were also made to the raw mill system. During wet weather when the moisture content of the raw mill product increased, the raw mill transfer line to the kiln feed silos would plug occasionally.

   Adding a small air compressor to the pneumatic pump and later changing the raw mill transfer line size helped mitigate this problem. However, the raw mill was never a bottleneck to the kiln, so these changes would have no impact on NOx. Even if the raw mill became a bottleneck to the kiln after the calciner improvements, and was subsequently debottlenecked by the compressor or transfer line changes, that debottlenecking would have no effect on NOx emissions. As discussed above, any additional kiln production enabled by the calciner improvements did not result in an increase in NOx emissions. Producing more raw feed to meet that additional kiln production capacity would also not increase NOx emissions.

   In addition, the raw mill system and the pyroprocessing system are not directly connected. There is a homogenizing silo and a kiln feed silo that each provide surge capacity for raw mill product storage. These two silos have a combined capacity of 5,700 tons, or 48 hours of kiln feed for the pyroprocessing system at the maximum permitted feed rate of 120 tph. The changes to the pump and convey lines were implemented to minimize plugging and the consequent maintenance effort required to clear the line, and due to this existing intervening storage capacity would not provide additional kiln feed capacity.

4. In the Walter L. Greer expert report for the United States, Mr. Greer attributes a "Tempering Tee" to debottlenecking the kiln. The author visited the Lyons plant on April 14, 2011 and, as part of the plant tour, examined the dryer and ductwork to the baghouse specifically to see the tempering tee and how it was intended to function. The Lyons plant personnel pointed out a rectangular steel plate that had been welded to the ductwork above the combustion gas outlet of the dryer and told me that was where the tempering tee had been installed. Immediately above this area was a louver bleed air damper that was functioning. The Lyons plant personnel explained that they had tried the tempering tee for several years and it did not function as expected so they welded a steel plate over it years ago. Apparently, Mr. Greer mistakenly thought the louver bleed air damper was the tempering tee. The bleed air damper was installed at the time of the dryer conversion from a roaster in about 1980. The function of the bleed air damper was to protect the dryer fabric filter bags from high temperatures and avoid severe damage to the bags in the event of a temperature spike. The tempering tee had no effect on the dryer production rate, the raw mill production rate or the clinker production rate. Accordingly, the tempering tee could not affect NOx emissions.

5. The finish mill system at the Lyons plant was also improved. However, because the finish mill is an independent downstream system that does not bottleneck the kiln, any changes to this system could not have affected NOx emissions.

6. An annulus boost fan was installed in 2000 to provide a higher tip velocity flame. Unstable flames in a rotary kiln – which were occurring at the Lyons plant – adversely affect kiln operations and are unsafe for plant equipment and personnel. An unstable flame generally has a variable ignition point or plume length and during kiln upset conditions can result in the flame going out, with an explosion resulting a few seconds later. Additional concerns with a burner pipe with a low tip velocity (e.g., less than 7000 fpm) is the possibility of the flame propagating back into the burner pipe and into the coal mill system resulting in a fire or potential explosion in the fuel preparation system. Lyons addressed these concerns through installation of an annulus boost fan. As a result, improved flame stability was achieved which improves kiln operational stability, which will in turn typically reduce NOx emissions.

7. In summary, the efficiency improvements to the calciner and the use of OE together resulted in an increase in clinker production but a decrease in annual NOx emissions. The ID fan motor change was a replacement of the existing ID fan motor that suffered a catastrophic failure. It was not planned as a project to increase clinker production and had a minimal effect on NOx emissions. Changes to the raw mill, tempering tee, and finish mill had no effect on NOx emissions. The annulus boost fan was added to improve combustion conditions in the rotary kiln and would have improved kiln operational stability, which would have reduced NOx.

2.0 GENERAL INFORMATION ABOUT THE PORTLAND CEMENT INDUSTRY

2.1 DESCRIPTION OF THE PROCESS

There are many variations possible when a new cement plant is designed. These variations will depend, to a great extent, on the raw materials economically available and on the final product that will be produced. Cement manufacturing facilities are generally separated into three major subgroups: raw material acquisition and preparation, pyroprocessing, and cement grinding and shipping.

Cement plants are designed with sufficient storage capacity for raw materials, clinker and finished cement in order to minimize the influence of one operating subsystem on the following subsystem(s). For example, a typical cement plant will have sufficient raw material storage for on-site materials for a minimum of four days of production. This allows the quarry to operate during the week and be off on weekends. The four day supply insures sufficient limestone for a long-weekend. Purchased raw material storage depends on the delivery schedule of the materials. Sufficient storage of purchased raw materials is provided to insure an on-site supply in the event that the outside supplier has delivery problems.

Raw materials that have been proportioned and ground into raw mix are stored in homogenizing and kiln feed silos. These silos are designed to isolate the raw material supply and the raw mill system from the pyroprocessing system. Storage for ground raw mix ranges from a one-day supply to as many as four or five days of supply for the pyroprocessing system. The amount of ground raw mix storage depends on the perceived reliability of the raw mill system and on the amount of raw mix inventory necessary to achieve the required homogenization.

Clinker and cement storage capacities are usually determined together. The total amount of clinker and cement storage will depend on the cement shipping seasonal variability. In areas of the country that have a high seasonality (i.e., ship 10 to 20 times as much per month during the construction season versus the winter), cement storage is usually determined so that the finish mill(s) can operate sufficient hours during the year to produce the cement necessary. There is a financial tradeoff between installing a finish mill that can produce at very high production rates with only modest cement storage and installing a finish mill that is not so large but has substantial cement storage. The clinker storage requirements will depend on the selection of the finish mill(s) production rate and the cement storage. If less cement storage is selected, then more clinker storage will be required.

The following is a description of a typical cement manufacturing facility.

2.1.1 Raw materials acquisition and preparation

Cement plants are typically located near an extensive source of limestone. Limestone is the most commonly used raw material and will usually represent between 80% and 95% of the total raw materials required. Other materials such as clay, shale, sand, iron ore, coal-fired power plant ash, and other available materials make up the remaining 5% to 20% of the raw material requirements.

Limestone is quarried by either drilling and blasting the stone or, if the material is soft enough, by ripping it with a large bulldozer. The limestone is conveyed to the primary crusher by large quarry trucks where the raw material size is reduced from one meter edge length or more to less than 100 mm edge length. A secondary crusher may also be used. The use of a secondary crusher will depend on the raw material characteristics, desired crushing rate (i.e., tons per hour), the required crushed stone product size, and several other factors.

The crushed raw materials are typically proportioned together in a blended bed storage pile(s) that provide(s) at least a 4-day stockpile of blended raw materials. The blended storage pile(s) can be either longitudinal or circular piles with the selection depending on raw material variations, future plans for increasing production, initial capital cost, and other factors. The average chemistry of the raw materials placed in the blended bed is typically determined by an on-line analyzer that provides real time chemical analyses of the materials as they are placed on the blended bed storage piles.

The blended raw materials are withdrawn from the blended storage piles and usually conveyed by rubber belt conveyor to a mill feed bin. Raw material additive bins and an on-line chemical analyzer are typically located before the raw mill to provide final chemical correction to the raw materials before they are dried and pulverized in the raw mill.

The raw mill is typically a vertical roller mill, although other types of mills can be used. Hot exhaust gases from the kiln preheater are ducted through the roller mill to provide the necessary energy to evaporate raw material moisture. Moisture concentrations of more than 20% can be dried in a vertical roller mill if special design considerations are used.

The dried (<1% moisture) and pulverized (80% smaller than 74 μm) raw materials are conveyed to homogenizing and storage silo(s). Until the 1990's it was typical to have a 3-day supply of dried and pulverized raw materials (kiln feed) in homogenizing and storage silos. Since the widespread adoption of on-line chemical analyzers for raw materials, current practice is to have 24 to 36 hours supply of kiln feed. In the past, the large store of kiln feed was required to achieve the necessary raw material blending. Use of on-line analyzers and more reliable mill systems have allowed new cement plants to reduce the quantity of kiln feed held in homogenizing silos because the raw material chemical uniformity from the raw mill has improved substantially. Kiln feed is withdrawn from the homogenizing and storage silo(s) at a carefully controlled rate and conveyed to the kiln system. Throughout the raw material acquisition and preparation systems, inventories of raw materials are maintained so that down time of a day or several days in one area of the system does not result in down time in subsequent systems.

2.1.2 Pyroprocessing

Rotary cement kilns are cylindrical, refractory-lined steel furnaces that range from 50 to over 215 meters long and average about 4.5 meters in diameter. They are inclined slightly so that from the high end, the kiln feed tumbles towards a high temperature (>1800°C) flame in the main combustion zone as the kiln slowly rotates. The flame is fueled by powdered coal, petroleum coke, natural gas, oil, recycled materials, or other alternative fuels. The extreme heat required by the process causes a portion of the raw materials to melt and to be transformed into lumps or balls called "clinker." The clinker passes under the flame and falls into the clinker cooler where air is passed through the clinker until the temperature is low enough that the clinker can be conveyed and placed into storage. A significant portion of the heated air from clinker cooling is returned to the kiln or calciner as preheated combustion air.

Each cement kiln falls into one of two major pyroprocess types, wet or dry. Plants using the wet process add water to the raw materials during grinding to produce a fluid kiln feed called a "slurry." The wet process simplifies mixing and proportioning of the raw materials to achieve a more uniform chemical composition, but the added water must be evaporated from the raw materials before the transformation into clinker can occur. To aid in the transfer of heat from the kiln gases to the slurry, metal chains are placed in the feed end of the kiln. The chains are heated and as the chains come in subsequent contact with the slurry, heat is transferred.

The dry process is divided into three different subtypes -- long dry kilns, preheater kilns, and preheater/calciner kilns. The long dry kiln is very similar to the wet kiln except that the raw materials are ground dry and enter the kiln as a dry powder instead of a slurry. Long dry kilns may also use kiln chains to transfer heat from the kiln gases to the kiln feed.

Preheater kilns also use dry kiln feed, but instead of the dry kiln feed directly entering the kiln, a preheater is first used to further dry and heat the kiln feed to calcining or near calcining temperatures. Typically, the preheater is a series of cyclone-type vessels where hot gases from the kiln rise through the preheater as the kiln feed falls in a countercurrent pattern down through the vessels. The heat transfer rate of a preheater kiln is significantly better than that of a long dry kiln.

A preheater/calciner kiln uses the preheater vessels to dry and heat the kiln feed, but an additional vessel is added between the preheater and the rotary kiln. This vessel, a calciner, allows fuel to be burned in a secondary combustion zone in the pyroprocess. The hot gases produced by combustion in the calciner come into intimate contact with the kiln feed and the raw materials calcine (thermally decompose calcium carbonate, $CaCO_3$) into calcium oxide, CaO, and carbon dioxide, CO_2. Because the heat transfer of a preheater/calciner kiln is significantly better than a long dry kiln, this "precalcining" step decreases the heat requirement of the entire pyroprocessing activity.

The initial process that occurs in a kiln is to remove moisture from the kiln feed. In dry process kilns the kiln feed may contain 0.5% to 1.0% moisture. In a wet process kiln the feed contains 30% to 44% moisture. Evaporating this moisture requires a minimum temperature (100°C at atmospheric pressure) and a heat input of 540 kcal per kg of water.

One of the first chemical reactions to take place in the pyroprocess occurs to the calcium carbonate ($CaCO_3$), which is the primary chemical in the raw materials used for cement making and is generally 75% to 80% of the kiln feed. After moisture has been removed, the feed must be heated to thermally decompose the $CaCO_3$ into CaO and CO_2. This chemical reaction, calcination, begins at a temperature of about 800°C, is completed at about 950°C, and requires a heat input of about 415 kcal/kg of $CaCO_3$.

The CaO is then chemically combined with other minerals in the kiln feed and transformed into clinker compounds. Formation of clinker compounds in the main combustion zone (burning zone) of a cement kiln is a net exothermic reaction and liberates about 100 kcal/kg clinker. These reactions require minimum initiation temperatures of from 1,200°C to about 1,450°C, depending on the chemical composition of the kiln feed. If the raw materials in the kiln are not heated to these minimum temperatures, the cement clinker compounds do not form. If the cement clinker compounds do not form, the exothermic heat of reaction is no longer available to the kiln system and the chemical processes stop. More importantly, the clinker produced will be of unacceptable quality to produce Portland cement.

Clinker produced by the pyroprocessing system is conveyed to storage for subsequent processing into the final product, cement. Clinker storage capacities vary substantially because of varying seasonality in shipping patterns between different geographically areas. For example, shipping patterns in the southern states are usually rather smooth because construction activities are not impacted by weather. Shipping patterns in northern states can vary substantially between summer and winter. The capacity of the clinker storage is based on the anticipated shipping patterns so that the size of the clinker inventory is sufficiently large that the seasonal shipping patterns do not affect the pyroprocessing system's operation.

2.1.3 Cement grinding and shipping

Clinker is withdrawn from storage and metered into the finish mills with, typically, gypsum and potentially other additives to produce the required product for the local market. Several different types of cement can be produced if market demand dictates. Cement storage is based on the number of different types of cement that might be produced and on the seasonal shipping patterns discussed above. Cement storage capacities are selected to minimize the impact of finish mill operations on operation of the pyroprocessing system.

2.2 COMPOSITION REQUIREMENTS

Portland cement is a unique combination of three major and numerous minor compounds.

A typical clinker will have a chemical composition within the following ranges for each particular compound:

Calcium Oxide or Lime	CaO	60% - 67% by weight
Silica	SiO_2	18% - 25%
Alumina	Al_2O_3	3% - 8%
Iron Oxide	$Fe_2 0_3$	2% - 6%
Magnesium Oxide	MgO	0% - 6%
Sulfur Trioxide	SO_3	0% - 2.0%

Clinker may also contain other minor compounds, generally less than 1% each, of potassium oxide ($K_2 0$), sodium oxide ($Na_2 0$), phosphorus pentoxide (P_2O_5), titanium oxide (TiO_2), and manganese oxide ($Mn_2 0_3$). Generally, all the above-mentioned compounds are determined and expressed as oxides.

Clinker is subsequently ground with a small quantity of gypsum (needed to control the setting time of concrete) or other additives into the extremely fine (90% $< 44\mu$) gray powder known as Portland cement. Cement is ground in ball mills although some cement grinding systems use high pressure grinding rolls as a pregrinding unit. Vertical roller mills are becoming more prevalent for grinding cement because of their lower energy consumption per ton of cement. The final product is then conveyed to storage (typically silos) and shipped, as needed, to customers.

2.3 CEMENT TYPES

Cement is a finely pulverized mixture of cement clinker and some form of calcium sulfate, usually gypsum. Cement clinker is an intermediate product composed of various calcium silicates, calcium aluminates and calcium ferrites and is manufactured in a rotary cement kiln.

There are several different types of cement in common use and a great number of different specifications may apply to selected cement types. In addition to cements used as construction materials, cements are also manufactured for other uses, such as oilwell cements and high alumina cements used as high temperature refractory. A brief summary of typical ASTM specifications (ASTM C-150) for various cement types in the United States follows:

ASTM Cement Type	Description
I	General Purpose
II	Moderate sulfate resistance and Moderate heat of hydration
III	High early strength
IV	Low heat of hydration (very rare)
V	High sulfate resistance
Optional Specifications:	
A	Air Entraining
LA	Low alkali

2.4 QUARRY OPERATIONS

In the normal quarrying operation, the quarry face material could be as large as 80" (2000 mm). These raw materials are then crushed to the required mill feed size in a primary crusher. Some plants are fortunate to achieve the necessary size reduction in one stage of crushing, but occasionally two stages of crushing are required.

Crushing and Milling generally refer to processes, which may be differentiated by the size of the end product from each operation. Milling is the further size reduction of the raw materials to a powder after crushing. Milling could further be applied to the grinding of cement. This is commonly referred to as Finish Grinding or Cement Grinding.

Whether the raw materials are ripped with a bulldozer or blasted with high explosives, crushing is a relatively simple procedure. Large rocks are to be made into smaller rocks. Apart from the feed size consideration, wear and stickiness are the two main considerations. Relatively large slabs of stone need to be crushed to a size consistent with the type of raw mill being used. Where ball mills are used as raw mills, the final feed size typically needs to be 100% passing 1" (25 mm), and 80% passing 1/2" (13 mm). For a roller mill, the feed size is typically 3" (75 mm) or less, the top size being dependent on the diameter of the grinding rollers.

2.5 RAW MATERIAL PROPORTIONING AND BLENDING

Stable kiln operation requires that addition rate of materials and fuel entering the kiln be held as constant as possible and the kiln feed chemistry must also be as uniform as possible. The chemical variability of the raw materials that will be used must be evaluated to determine the proper raw material proportioning and blending system to use. The degree of work necessary to achieve uniform kiln feed depends on the variability of the raw materials and the number of raw materials.

The objective of the quarry, raw material storage areas, raw mill, and a major concern of the laboratory is the proper preparation of kiln feed. The production of a quality product, cement, must begin at the quarry face. Proper selection of raw materials for their chemical composition and the blending of these raw materials should be a continuing process from the quarry to the feed end of the kiln.

Cement strengths (i.e., 28-day compressive strengths) are significantly affected by the operation of the kiln and the chemistry of the clinker and to a lesser degree by the operation of the finish mills.

Preparation of kiln feed can have significant effects on the operation of a kiln. Kiln feed chemistry has significant effects on the fuel efficiency of the kiln, the production rate of the kiln, the strength gain characteristics of the cement produced from the clinker, and the energy required to grind cement among other things. Kiln feed chemistry can also affect emissions of NOx and SO_2. The cement industry characterizes kiln feed as "hard-to-burn" or "easy-to-burn." The difference between the two can result in specific fuel consumption increases of 0.5 mmBtu/ton clinker with NOx emissions increasing substantially. Kiln feed chemistry at Lyons changes frequently due to changing clinker types, changes in available raw materials (e.g., use of fly ash), changes to reduce CKD land-filling, and other things.

2.6 KILN FEED PREPARATION

Raw materials are proportioned from blended storage piles, raw material feed bins, or a combination of both into the raw mill. These materials are dried and pulverized into a fine powder (typically 80% < 74µ). The raw mill system will contain an air separator that classifies the material according to size. Material that is sufficiently pulverized, or ground, will be removed from the system and conveyed to homogenizing and storage silo(s). The oversized material is returned to the raw mill for additional grinding.

The required kiln feed fineness is determined by the type of raw materials and how the kiln operates with slightly different raw material fineness levels. To minimize power consumption in grinding, the coarsest raw mix fineness consistent with good clinker quality is desired. However, if the raw mill production rate is adequate, reducing raw mix fineness can provide false economies. Finer raw materials react easier and more completely in the kiln. If a raw material used as a source of silica contains coarse quartz, such as sand, coarse kiln feed can result in burning problems with the kiln and poor quality clinker.

Homogenizing and kiln feed storage silo(s) are sized for the degree of homogenization required to achieve the desired degree of chemical uniformity and for the necessary inventory to permit maintenance on the raw mill system and associated equipment without affecting kiln operations.

Chemically uniform kiln feed is withdrawn from the kiln feed silo at a precisely controlled rate and conveyed to the kiln system for further processing.

2.7 DIFFERENT RAW MILL TYPES

The two primary types of raw mills are ball mills and vertical roller mills. In addition to these two, several high pressure grinding roll systems have been installed to dry and grind raw materials. Vertical roller mills tend to have lower energy consumption per ton of raw materials and also offer the ability to use waste heat from the kiln preheater to dry the raw materials as they are ground.

2.7.1 Ball Mills

Much simplified, a ball mill is a horizontal cylinder, partially filled with steel balls, that rotates to produce a tumbling action in the ball charge. Raw mills grind raw materials to the required fineness (typically about 80% smaller than 74μ).

The methods for drying raw materials associated with the operation of ball mills generally fall in one of the following three groups:

1. Pre-drying ahead of the mill
2. Drying within the air separator
3. Combined drying and grinding within the mill

2.7.2 Vertical Roller Mills

Roller mills have a rotating table on which 2, 3, or 4 rollers grind raw materials. The weight of the rollers and hydraulic pressure provide the grinding force to pulverize the raw materials. Hot gases are passed through the mill for drying and material conveying. Ground raw materials are forced to the outer edge of the table by the table rotation. The hot gases flow around the table edge from below and entrain the pulverized raw material.

The gas stream with the pulverized raw material passes through an air separator. The fine material continues out of the roller mill and is subsequently conveyed to kiln feed storage. The oversized material is returned to the rotating table for additional grinding.

2.8 DIFFERENT KILN TYPES

Kilns are normally separated into two primary classes, wet process and dry process. The feed for a wet process kiln is mixed with water to form a slurry with 30 to 45% moisture content. The feed for dry process kilns is dry (≤ 1% moisture). Dry process kilns are further subdivided into long dry, preheater, and calciner kilns.

Because dry process kilns are substantially more fuel efficient, very few new, wet process kilns have been built in the world in the last 20 years. Wet kilns typically have an enlarged feed end and have chains installed in the first 20 to 25% of the kiln length to recover heat from the kiln gases.

Long dry kilns are similar to long wet kilns except the raw materials are fed into the kiln dry rather than as a slurry. Long dry kilns also typically have kiln chains installed to recover heat from the kiln gases and transfer it to the incoming feed.

A typical preheater kiln has several (4 to 6) cyclone stages to preheat the incoming feed before it enters the kiln. The fresh feed is placed into the gas duct at the top of the preheater string and flows generally countercurrent to the hot kiln gases. The raw material feed is heated from ambient to about 1600°F to 1700°F in a preheater kiln in a few minutes, as opposed to a long kiln that may take several hours to heat the fresh feed to these temperatures.

A calciner kiln also has several preheater vessels, which rapidly and efficiently transfer heat from the hot kiln gases to the raw material feed. In addition to the preheater vessels, a calciner kiln typically burns about 50 to 60% of the total fuel requirements in the calciner vessel in the preheater tower as opposed to a preheater kiln, which burns all of the fuel in the burning zone. The largest energy requirement in a kiln system is for calcining limestone. A preheater kiln will typically calcine about 30 to 40% of the limestone before the kiln feed enters the kiln. A calciner kiln adds additional fuel at the bottom of the preheater, directly where it is needed, to accomplish nearly all (90 to 95%) of the calcining before the feed enters the kiln. This dramatically reduces the thermal load on the burning zone and the amount of gases exhausted from the rotary kiln.

A calciner is typically dimensioned using the following parameters:

Kiln type
Fuel type to the calciner
Fuel flow to the calciner
Retention time of gas/material
Gas velocity in the vessel
Gas velocity at the calciner inlet

2.9 CLINKER COOLERS

A clinker cooler is used to cool the clinker for subsequent handling and to recover heat from clinker by preheating combustion air (secondary and tertiary air) for the kiln and to provide hot gases for drying other materials such as coal or high moisture raw materials.

One of the first requirements to maximize heat recuperation is to supply the clinker cooler with the proper amount of air for the amount of clinker. If too much air is supplied by the undergrate fans, it will cool the secondary and tertiary air and overload the vent fan. If too little undergrate air is supplied, the clinker discharge temperature will be too high and it will risk damage to the grates because of high temperature. In general, it takes about 2.5 to 3.5 kg of air to cool 1 kg of clinker.

The clinker temperature at the cooler inlet is about the melting point of the metals of which the cooler is constructed. Undergrate fan air flow must be maintained at all times to avoid severe damage to the equipment. No air flow or insufficient air flow will quickly result in damage to the clinker cooler.

Heated secondary air from the clinker cooler typically supplies 12 to 20% of the total process heat required by the kiln. Secondary air temperatures range from 1000°F on wet process kilns to as high as 2000°F on calciner kilns.

Some kilns fire fuel in the riser duct that comes from the kiln exhaust to the preheater vessels. The amount of fuel fired in a riser duct by using combustion air from the rotary kiln is typically limited to about 20% of the total fuel requirement of the pyroprocessing system. The excess combustion air that was drawn through the kiln tended to cool the flame in the kiln. At firing rates of more than about 20% in the riser, the kiln flame became too cool to produce quality clinker. Calciner kilns firing 20% to more than 60% of the total fuel requirements in the calciner must have a combustion air supply that does not have to come through the kiln because of the cooling effect on the kiln flame. This is accomplished by installing a separate duct (called a tertiary air duct) to supply heated air from the clinker cooler for combustion in the calciner.

Proper clinker cooler operation in important because improved heat recovery from the clinker cooler will result in better fuel efficiency.[1] Also, proper clinker cooler operation in important because improved heat recovery from the clinker cooler will result in better fuel efficiency[2] which tends to reduce NOx emissions.

2.10 CEMENT MILLS

There are three primary factors that determine the power required to grind a given material. These factors are: Feed Size - larger feed size will increase power requirements; Product Size - finer grinding will increase power requirements; and Work Index - materials with a greater Work Index are harder to grind and will increase power requirements.

Mill performance depends not only on mill power considerations and mill system parameters, but to a great extent it also depends on the design, operation, and maintenance of mill auxiliary systems such as:

1. Air separators
2. Elevators
3. Dust collectors

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[1] USEPA, Alternative Control Techniques Document Update – NOx Emissions from New Cement Kilns, EPA-453/R-07-006. November 2007, p. 34.

[2] USEPA, Alternative Control Techniques Document Update – NOx Emissions from New Cement Kilns, EPA-453/R-07-006. November 2007, p. 34.

4. Air slides and screw conveyors
5. Cement coolers and cyclones
6. Pneumatic pumps and air compressors

2.11 AIR SEPARATORS

2.11.1 Conventional Air Separators

In a conventional mechanical air separator, the rejects that return to the mill contain substantial amounts of material fine enough to be finished product. There are several inefficiencies that can occur in conventional air separators, which usually arise for essentially two reasons:

1. A clean cut based on particle size is not made as the material drops off the feed plate. As a result, coarse material can be carried up and over and exits with the fines. In addition, fine material can be trapped by the descending flow of rejects and exits with the rejects back to the mill.

2. The circulation of air back in through the return air vanes carries substantial amounts of fine material that again can be trapped by the descending umbrella of rejects.

2.11.2 High Efficiency Air Separators

The principles of operation of high efficiency separators differ from conventional separators as follows:

1. Air vented from the separator is withdrawn by an external (dust collector) fan.

2. There is no internal circulation of air. Air is passed through once.

3. All the fines (product) are carried out with the separator vent air.

Most of the inefficiencies of the conventional air separator are eliminated when compared to a high efficiency separator.

2.12 CLINKER FORMATION

Kiln feed for clinker contains five major oxides and several trace elements. The five major oxides are SiO_2, Al_2O_3, Fe_2O_3, CaO and MgO. The trace elements, are SO_3, Na_2O, K_2O and Cl. Only SiO_2, Al_2O_3, Fe_2O_3, and CaO take part in the clinker reactions.

Although SiO_2, Al_2O_3 and Fe_2O_3 are normally present in raw materials as oxides, CaO and MgO require most of the heat used for raw material preparation. CaO is formed in the kiln from $CaCO_3$ (limestone) and MgO is formed in the kiln from $MgCO_3$ (dolomite) according to the following chemical reactions:

$$CaCO_3 \rightarrow CaO + CO_2\uparrow$$
$$MgCO_3 \rightarrow MgO + CO_2\uparrow$$

It is the release of CO_2 (carbon dioxide) from limestone and dolomite that requires most of the heat for preparing raw materials for reaction.

2.13 FUEL COMBUSTION

There are several steps that any fuel undergoes as it is combusted (oxidized or burned). First, the molecules of the fuel are heated. Then hydrocarbons are cracked or decomposed which liberates hydrogen and the hydrogen burns, creating H_2O or water. The free carbon either burns and forms CO and CO_2 or soot particles are formed. The soot particles burn and form CO and CO_2. The carbon monoxide is burned and forms CO_2.

The requirements for these combustion reactions to occur are sufficient heat to dissociate the hydrocarbon molecules and excess oxygen to complete the combustion reactions. The hydrogen and carbon particles must be given the chance to meet and combine with oxygen as soon as possible. This is achieved by rapidly mixing the fuel and air. If oxygen is in exactly the right amount (termed stoichiometric amount), the chances of every hydrogen and carbon molecule finding an oxygen molecule are very low. Therefore, excess oxygen must always be provided. This will help prevent the presence of unburned fuel in the exhaust gases.

2.13.1 Endothermic and Exothermic Reactions in the Kiln System

The major endothermic reactions that occur in the kiln are evaporation of moisture (from fuel and feed) and feed calcination (release Of CO_2), as follows:

$H_2O_{liq} \rightarrow H_2O_{vapor}$	(+ 970 Btu/lb of H_2O @ 212°F)
$CaCO_3 \rightarrow CaO + CO_2\uparrow$	(+ 780 Btu/lb of $CaCO_3$ @ 1650°F)
	(+ 1400 Btu/lb of CaO)
$MgCO_3 \rightarrow MgO + CO_2\uparrow$	(+ 510 Btu/lb of $MgCO_3$ @ 1100°F)
	(+ 1100 Btu/lb of MgO)

Other minor reactions occur as the feed progresses through the kiln toward the burning zone; e.g., the removal of the combined water from kaolin (clay) in the raw feed.

Major exothermic reactions that take place in the kiln, other than fuel combustion, is the reaction between lime, silica, alumina and iron oxide to form the clinker compounds C_4AF, C_3A, C_2S, and C_3S. These reactions occur at temperatures of 1200 to 1450°C and release heat equivalent to 100 kcal/kg of clinker produced.

2.13.2 Heat Transfer

The primary purpose of the pyroprocessing system is to transfer heat from the fuel and other heat sources to the raw material. Heat transfer occurs by one of three mechanisms.[3]

Convection is heat transfer from one material to another through fluid motion. For example, heat is transferred from the hot kiln gases to incoming feed in a preheater vessel. A formula for heat transfer by convection is:

$$Q = hA(T_1 - T_2)$$

Q	=	rate of heat transfer in Btu/hr.
h	=	coefficient of heat transfer in Btu/hr/sq.ft./°F
A	=	area available for heat transfer in sq.ft.
$T_1 - T_2$	=	temperature difference in °F

Conduction is heat transfer within a given material or heat transfer from one material to another in physical contact by transfer of vibrational energy from one molecule to another. For example, heat is transferred from hot kiln chain to the cooler kiln feed by conduction. A formula for heat transfer by conduction is:

$$Q = kA\,(dt/dx)$$

Q	=	rate of heat transfer in Btu/hr.
k	=	thermal conductivity in Btu/hr/ft/°F
A	=	area available for heat transfer in sq.ft.
dt/dx	=	temperature gradient in °F/ft

Radiation is the transfer of heat from one material that is not in contact with another. Heat transfer by radiation is an important factor within the kiln. The flame, the refractory and the coating radiate heat to the feed in the kiln. A formula for heat transfer by radiation is:

$$Q = \sigma\varepsilon A\,(T_F^4 - T_P^4)$$

Q	=	rate of heat transfer in Btu/hr.
σ	=	Steffan-constant in Btu/hr/sq.ft./°F
ε	=	emissivity (0.0 – 1.0)
A	=	area available for heat transfer in sq.ft.
T	=	surface temperature in °R (°R = °F + 460)

[3] See Young, Gerald L and F. M. Miller, Ph.D., Chapter 3.3 Kiln System Operations in Cement Manufacturing, *Innovations in Portland Cement Manufacturing*, Portland Cement Association, p. 310-311.

2.14 COMBUSTION CONTROL

Combustion control is critical for safe and efficient operation of a kiln. Oxygen for combustion can come from several sources, such as:

- Primary air from burner pipe blower
- Hot secondary and tertiary air from the clinker cooler
- Air leaks around the firing hood
- O_2 lances or other sources of O_2

The "correct" or proper amount of air for combustion is generally determined by an oxygen analyzer at the exit of the kiln and the outlet of the preheater. Oxygen levels should be controlled so that sufficient excess oxygen is available to insure all of the fuel is completely burned.

If insufficient air is drawn through the kiln, incomplete combustion occurs in the burning zone and combustible gases remain in the exhaust. This is inefficient because the heat value of the unburned fuel is not available for use in producing clinker. When the oxygen supplied to the kiln approaches the level of oxygen that is exactly the required amount for the fuel supplied (this is termed stoichiometric levels), the flame will become very long. If combustible gases leave the preheater and encounter an air leak that supplies oxygen, rapid combustion or explosions can occur. This is the reason that a combustibles or CO analyzer is used in addition to the oxygen analyzer.

It is also inefficient to use too much excess air. All of the excess air must be heated to the operating temperature of the pyroprocessing system, which requires a large fuel input just to heat the air. Since oxygen is only about 21% of air, each 1% O_2 indicates that about 5% excess air is being drawn through the kiln.

2.15 FAN PERFORMANCE

A pyroprocessing system uses large fans to move enormous amounts of air or gases into and through the pyroprocessing system. For example, in a modern, efficient dry process cement plant, the total mass air flow into the clinker cooler is about three times the mass of clinker into the cooler. That is, if a kiln is producing 100 tons per hour there is about 300 tons per hour of air being blown into the clinker cooler to cool the clinker. The ID fan will move more than twice the mass of gases as the clinker production rate. If a pyroprocessing system is producing 100 tons per hour, the ID fan will be moving more than 200 tons per hour of gases.

Each fan has a specific set of conditions describing how that fan is expected to perform in the field. These conditions include the following:

Airflow (cfm) at the fan inlet.
Static Pressure (inches H_20).
Power Input (kW).

Fan Speed (RPM).
Air Density (lb/cu ft) at fan inlet.
Fan Efficiency (%).

Fan speed refers to the speed of rotation in revolutions per minute (RPM) of the fan shaft. Changing fan speed will change fan cfm, static pressure (SP) and brake horsepower (BHP) according to the following relationships:

$$\text{New cfm} \div \text{Old cfm} = (\text{New RPM} \div \text{Old RPM})$$
$$\text{New SP} \div \text{Old SP} = (\text{New RPM} \div \text{Old RPM})^2$$
$$\text{New BHP} \div \text{Old BHP} = (\text{New RPM} \div \text{Old RPM})^3$$

Therefore doubling the speed of the fan will double the volume at the fan, but will increase SP by a factor of 4 and will increase BHP by a factor of 8.

2.16 KILN OPERATION

Stable kiln operation requires that addition rate of materials and fuel entering the kiln be held as constant as possible. The kiln feed chemistry must also be as uniform as possible. If all of these parameters are held perfectly constant, the kiln would be well behaved and no corrective control would be necessary from the operator once the proper operating conditions were established. However, none of these parameters can be held perfectly constant.

In order to understand the function of the clinker manufacturing equipment it may be helpful to review the steps of the clinker formation process.

- The first reaction that occurs after feed is placed in the kiln is drying water.

<u>30°C – 900°C</u>

30 - 100	Evaporate water (free water)
100 - 300	Evaporate water of crystallization
400 - 900	Remove chemical water

- After the moisture is evaporated, kiln feed is heated and carbon dioxide is driven off of calcium and magnesium carbonate.

<u>600°C – 1100°C</u>

600 - 900	$CaCO_3 \rightarrow CaO + CO_2$ Limestone decomposes into carbon dioxide and lime (CaO). Lime is the main reactive component that produces clinker.

800 The first clinker formation reactions begin. Formation of ferrites, aluminates, and belites (C_2S) begins and continues through temperatures of 1100°C

After lime has begun to form, and the temperature of the material increases further, the sintering or clinkering reactions begin to occur in the burning zone:

<u>1100°C – 1450°C</u>

1250°C Formation of the liquid phase (Al_2O_3, Fe_2O_3) and begin formation of alite (C_3S).

1450°C Final formation of alite and belite.

Once the reactions are completed, the clinker begins to cool in the last few meters of the kiln.

<u>1450°C – 1250°C</u>

1300°C – 1250°C Crystallization of aluminates and ferrites.

Gas flow is counter-current to material flow. Gases enter the kiln as secondary air at about 700 to 1000°C and rise very abruptly to over 1800°C as the heat from fuel combustion and the exothermic heat of clinker formation are released. The gases gradually transfer heat to the material in the kiln as they travel back through the kiln. This heats the material and cools the gases. The gases then enters the preheater at more than 900°C and finally exit the preheater at about 350 to 400°C. Throughout the balance of the system, heat is transferred from gas to material.

3.0 THE LYONS PLANT AND ITS COMPARISON TO A TYPICAL PORTLAND CEMENT PLANT

When the Lyons plant was originally constructed in 1969, it consisted of a quarry, primary crushing, raw material dryer (shale roaster), a secondary crusher, a raw ball mill, kiln feed homogenizing and storage, a long dry process kiln with a reciprocating grate clinker cooler, clinker storage, a finish ball mill, and cement storage silos. In 1969, this was typical for the design of a Portland cement plant.

The raw materials at the Lyons plant consist of limestone and shale from an on-site quarry, purchased iron ore, fly ash, and other raw materials. The shale contains kerogens, or is oil-impregnated. Raw materials from the quarry have relatively high moisture levels so a rotary dryer was included in the original plant design to pre-dry the raw materials. The dried raw materials are sent from the dryer to a secondary crusher and then to the raw material feed bins. The raw mill does not have additional drying capacity.

Throughout the Lyons plant, various independent material storage silos and designated areas have been allocated to facilitate the operation of the plant by minimizing the influence of one operating subsystem on a subsequent operating subsystem. Inventory capacities are selected at each stage to optimize plant performance and minimize downtime. For example, raw materials are quarried, conveyed to the plant, and stockpiled for use in the raw mill system. The dried, crushed raw materials are stored in mill feed bins where they are carefully proportioned into the raw mill to be ground to the proper fineness. The ground raw materials (raw mix or kiln feed) are then stored in silos that have two full days of inventory for the pyroprocessing system. The kiln feed is withdrawn at a carefully controlled rate and conveyed into the preheater.

The pyroprocessing system at Lyons consists of a one-stage preheater, a calciner, a rotary kiln and a clinker cooler. Clinker from the pyroprocessing system is stored in an A-frame building with a capacity of 60,000 tons of clinker. An additional 120,000 tons of clinker can be stockpiled outside. Clinker and gypsum are conveyed to the finish mill system for processing into cement, which is stored in cement silos for subsequent shipment.

3.1 RAW MATERIALS

The primary raw materials used at Lyons are limestone and shale from an on-site quarry. Additional outside materials are also used including iron ore, silica, fly ash and other materials.

Raw material stockpiles separate the quarry operations from the raw mill system so that variations in quarry operations have minimal effect on operations of the raw mill system.

3.2 RAW MATERIAL BLENDING

The raw materials at Lyons are preblended by stacking them together with a radial stacker. They are then reclaimed by a front end loader and fed to a surge silo and then to the rotary dryer. The materials are dried in a natural gas fired rotary dryer and conveyed to a secondary crusher and then to the raw material feed bins. From the feed bins, the raw materials are proportioned together and fed into the raw mill system. Samples of the raw mill product are obtained and these samples are chemically analyzed with an X-ray fluorescence (XRF) which provides rapid chemical analyses (e.g., within a few minutes). The raw material feeders are adjusted periodically to produce the desired chemical composition for the kiln feed.

This is a typical design for a cement plant of the vintage of the Lyons plant. Today, preblended stockpiles with an on-line analyzer(s) would be typical.

3.3 RAW MATERIAL GRINDING

The Lyons plant has a separate natural gas fired raw material dryer to pre-dry the raw materials prior to being conveyed to the raw material feed bins. This is not a typical process since most cement plants use the raw mill to simultaneously dry and grind raw materials; however, some plants require a pre-drying step if the raw materials have high moisture contents.

The grinding mill at Lyons is a ball mill with an air separator which was the typical raw mill when the Lyons plant was constructed. The fine product from the separator is conveyed to a pneumatic pump (FK pump) and transferred via pipeline to the homogenizing silos.

In a typical cement plant constructed in the last 10 years, a vertical roller mill would have been selected to simultaneously dry and grind raw materials using preheater gases as the energy source to dry the raw materials.

3.4 RAW MIX BLENDING AND STORAGE

At the Lyons plant, proportioned raw materials from the raw mill pump undergo additional blending in a homogenizing silo. Kiln feed chemical uniformity is critical to smooth, efficient operation of the pyroprocessing system. The homogenizing and kiln feed silos provide blending and kiln feed storage. The importance of chemically uniform kiln feed is demonstrated by the following quotation by Con G. Manias:

> The difference between well homogenized and poorly homogenized kiln feed can be an increase of up to 200°C in the operating temperature of the

> burning zone, the associated high fuel consumption, high NOx, poor refractory life, and poor product reactivity. [4]

Kiln feed storage provides a stockpile of material to separate operation of the raw mill system from the pyroprocessing system. It allows time to shut down the raw mill system for routine maintenance and for unanticipated breakdowns without impacting operation of the pyroprocessing system.

The kiln feed blending and storage silos at Lyons are typical of cement plants constructed in the late 1960's and early 1970's.

3.5 PYROPROCESSING SYSTEM

In 1980, Fuller Company was selected to convert the Lyons kiln to a one-stage preheater, calciner kiln to mitigate and reduce hydrocarbon emissions.[5] A one stage preheater with a calciner was added to the kiln and the long dry kiln was cut approximately in half.

When kiln feed containing kerogens is heated slowly, as in a long dry kiln, the kerogens are boiled off and are emitted as unburned hydrocarbons. By converting the Lyons kiln to a one-stage preheater with a calciner, the feed is introduced into a high temperature environment (e.g., greater than 850°C or 1560°F) and the kerogens are burned rather than being emitted as unburned hydrocarbons.[6] The hot gases and feed are pulled through the calciner into a one-stage cyclone where they are separated. The feed, which is now 70 to 75% calcined, is then fed into the rotary kiln.

Combustion in a calciner is substantially different than combustion in the rotary kiln. In a calciner, no defined flame exists. Fuel is oxidized under "flameless" conditions. As with any combustion, good air fuel mixing is required for good combustion.[7] Because of the short residence time in the Lyons calciner and low temperatures (relative to the rotary kiln), achieving complete combustion can be difficult.

At the Lyons facility a problematic condition used to occur in which fuel fired into the calciner does not fully combust. Instead, the fuel is carried by the gases and continues to combust downstream of the calciner in the cyclone, and even at times in the conditioning tower following the cyclone. This condition is sometimes referred to as "overburn" and was the reason why the internal geometry of the calciner was modified in 1997. It is also part of the reason why CEMEX decided to inject oxygen into the calciner.[8] As discussed in greater detail in Section 4.1.1, these calciner efficiency improvements resulted in

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[4] Manias, Con G., Chapter 3.1, Kiln Burning Systems, Innovations in Portland Cement Manufacturing, Portland Cement Association, p. 261.
[5] Tseng, Herman and John W. Lohr, Oxygen Enrichment, *International Cement Review*, May 2001, CMXLYO00045538.
[6] Tseng, Herman and John W. Lohr, Oxygen Enrichment, *International Cement Review*, May 2001, CMXLYO00045538.
[7] Manias, Con G., Chapter 3.1, Kiln Burning Systems, *Innovations in Portland Cement Manufacturing*, Portland Cement Association, p. 258.
[8] Deposition of Randal Wiley, November 10, 2010, pp. 109, 110.

reducing carbon monoxide (CO) emissions, improved pyroprocessing fuel efficiency, and thereby reduced overall NOx emissions.

The Lyons plant has raw materials that tend to be high in alkali (potassium and sodium) which is deleterious to cement quality if they exceed certain values and are in the presence of reactive aggregates. When the Lyons plant was modified in 1980, an alkali bypass was installed so that, when necessary, cement kiln alkali dust could be removed from the system and landfilled to lower the clinker alkali concentration to meet the customer demand for Low Alkali specification cement (cement with less than 0.60% alkali reported as Na_2O). However, as the customer demands for low alkali cement vary, the amount of cement kiln dust (CKD) removed and landfilled varies with that demand. The Lyons plant considers landfilling CKD as undesirable and therefore developed recycling processes to allow CKD to be returned to the pyroprocessing system. By recycling CKD back to the pyroprocessing system, the overall energy efficiency of the system improves.[9]

As the kiln feed progresses through the kiln, it undergoes progressive chemical changes ultimately being heated to temperatures between 2650 to 2750°F which forms the clinker minerals. The clinker is then discharged into the clinker cooler.

3.6 FUEL FIRING SYSTEM

There are several different types of solid fuel firing systems and these are typically classified by how the pulverized fuel is handled after it leaves the grinding mill. The safest and simplest system is the direct fired system where raw fuel is fed in the mill where it is ground and dried. Hot air from the clinker cooler or gases from the preheater are pulled through the mill and are used to dry and convey the pulverized fuel. The pulverized fuel is then withdrawn from the mill with the mill fan which then blows the pulverized fuel directly into the combustion vessel. An indirect system is identical to a direct fired system up to the point where the pulverized fuel leaves the mill. In an indirect system, the gases stream and entrained pulverized fuel is passed through a dust collection device(s) and the moisture laden gases are vented to the atmosphere. The pulverized fuel is collected in a storage bin and metered into a pneumatic pump(s) to be conveyed to the combustion vessel(s).[10]

When the pyroprocessing system at Lyons was changed to a one-stage preheater kiln with a calciner in 1980, the coal firing system was an indirect-fired system with a multichannel burner pipe. In the original indirect system, the pulverized coal and air from the coal mill passed through a cyclone then to a baghouse. The plant had two fuel explosions in the new system and subsequently had it converted to a semi-indirect

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[9] See USEPA, Alternative Control Techniques Document – NOx Emissions from Cement Manufacturing. EPA-453/R-94-004, March 1994, p. 5-4 ("Recycling cement kiln dust from the dust collectors would reduce the energy requirement per ton of clinker. By increasing the thermal efficiency, NOx emissions are likely to be reduced per ton of clinker produced.").

[10] Young, Gerald L and F. M. Miller, Ph.D., Chapter 3.3 Kiln System Operations in Cement Manufacturing, *Innovations in Portland Cement Manufacturing*, Portland Cement Association, p. 313.

system. In the semi-indirect system the original baghouse was not used and the air exhausted from the cyclone was routed to the swirl chamber of the calciner. The Lyons staff found that when the cyclone performance deteriorated, excessive amounts of pulverized coal were exhausting with the air from the cyclone and an uncontrolled amount of fuel was being conveyed to the calciner. In the early 1990's, two smaller Raymond coal mills were purchased and installed as direct fired systems. One direct-fired coal mill system for the calciner and one for the rotary kiln. By using two separate coal mills the Lyons plant was able to fire higher fineness coal (85% passing a 200 mesh sieve) into the calciner relative to the coal fired into the kiln (70% passing a 200 mesh sieve) to reduce the burnover issues.[11]

The Lyons plant currently injects a portion of the fuel for the calciner into the riser from the kiln to the calciner. This is the operating principle of modern low-NOx calciners. Burning a portion of the fuel in the riser at Lyons will reduce overall NOx emissions similar to modern low-NOx calciners.

3.7 CLINKER COOLER

There are several types of clinker coolers, but reciprocating grate coolers, as used at Lyons, are the most common type, especially for cement plants built in the 1960s to 1990s. The grates are steel plates with multiple perforations. The holes are about the diameter of a wooden pencil. Alternating rows of grates move back and forth resulting in the clinker resting on the grates moving slowly through the clinker cooler.

Reciprocating grate clinker coolers originally operated with shallow bed depths (8 to 12 inches) of clinker, low static pressure fans, and relatively low retention times. These clinker coolers were typically capable of handling about three tons per day per square foot of cooler grate area (tpd/ft^2). As production increased on kilns, bed depths were increased (12 to 16 inches), medium pressure fans were installed, and clinker retention time was increased. With these changes, a cooler would typically be capable of handling 3.5 tpd/ft^2. If the retention time is further increased (resulting in clinker bed depths of 18 to 24 inches), and high static pressure fans are installed, a regular reciprocating grate cooler can handle about 4 tpd/ft^2. This is generally considered the maximum throughput for these types of coolers. Air beam or direct aeration type clinker coolers can operate with 30 to 36 inch bed depths and can handle about 5 tpd/ft^2.

In clinker coolers the vintage of the Lyons plant, the drag chain(s) were internal to the undergrate compartments and air leaks between compartments were serious problems in some cases. As the clinker moves from the hotter areas to the cooler areas, the air pressure in the compartments is progressively lower. If the drag chain penetrates the wall of the compartments, air leaks occur. This can result in reduced cooling efficiency and reduced energy recuperation from the clinker cooler. For these reasons, the Lyons plant

[11] Information provided during interview of RD Smith by telephone on May 10, 2011.

made changes to the internal drag chains in the clinker cooler to improve the energy efficiency of the clinker cooler.

The clinker cooler at Lyons is typical of the coolers installed during the time period. It is not an air beam or direct aeration type clinker cooler although it has a static grate section at the kiln outlet drop point. Proper clinker cooler operation is important because improved heat recovery from the clinker cooler will result in better fuel efficiency[12] and better fuel efficiency tends to reduce NOx emissions.

3.8 CLINKER WEIGHING AND STORAGE

Clinker production is typically estimated by measuring the kiln feed as it is fed into the pyroprocessing system. This is an imprecise method of weighing clinker production because losses of bypass dust, cement kiln dust (CKD), coal ash content of the fuel, and other factors can influence the kiln feed to clinker ratio, or yield. For at least 40 years, cement plants have been trying to measure clinker production directly from the outlet of the clinker cooler without much success. The relatively high temperatures (average 100°C with excursions up to 450°C) present severe conditions for weighing devices. Most cement plants have metal conveyors from the clinker cooler to clinker storage because belt conveyors typically cannot tolerate the high temperatures. Weighing materials on a metal conveyor presents mechanical challenges in addition to the high temperatures. Cement plants typically adjust for the imprecision in weighing clinker by taking physical measurements of clinker inventory and adjusting clinker production accordingly.

Clinker storage capacity at the Lyons plant is large relative to other cement plants in the author's experience. Clinker can be stored in an A-frame building which has a capacity of 60,000 tons of clinker.[13] An additional 120,000 tons of clinker can be stockpiled outside for a total clinker storage capacity at Lyons of 180,000 tons.[14] Clinker storage capacities for cement plants are sized so that variations in cement shipments do not affect the pyroprocessing system. An excellent description of the use of clinker storage to separate pyroprocessing from cement grinding is provided in the Technical Review Document for Operating Permit 95OPBO082:[15]

> From an overall perspective, the manufacturing process may be viewed as two segments – clinker production and cement production. The clinker storage allows the two processes to operate at different production rates. During periods of low demand for cement, clinker is accumulated. If cement is in high demand, the clinker production can be supplemented by purchase of clinker form other

[12] USEPA, Alternative Control Techniques Document Update – NOx Emissions from New Cement Kilns, EPA-453/R-07-006. November 2007, p. 34.

[13] State of Colorado, Construction Permit No. 98-BO-0259, September 11, 1998, CEMEX_NEIC_00000251.

[14] CMXLYO00607481, Colorado Department of Health, Emission Permit No. 84BO369F, Dated September 18, 1986,

[15] US00054616, Technical Review Document for Operating Permit 95OPBO082, Cathy Rhodes, September, 1999.

sources. The overall result is the clinker production can operate at a rather steady rate, while the cement production can operate in response to the current of projected demands.

At Lyons, the clinker storage capacity effectively insulates the cement grinding segment from the pyroprocessing segment. Changes in cement grinding rates or hours of operation have little or no effect on pyroprocessing rates or hours of operation.

3.9 FINISH MILL

The finish mill at Lyons is a ball mill originally designed with two Raymond air separators. Ball mills have been used extensively for many years to grind both raw materials and finish cement. Cement grinding has undergone substantial improvements in energy efficiency over the last 30 years. For example, high efficiency separators have replaced the older first generation air separators, including at the Lyons plant, where such separators replaced the original Raymond separators. Several plants have installed high pressure grinding rolls to pregrind cement which is subsequently fed to a ball mill for final grinding. This type of hybrid system has reduced the energy required for grinding cement substantially. Most new cement plants are currently installing vertical roller mills to grind cement. They offer good efficiency with respect to specific power consumption and can grind cement at high output rates.

The original finish mill and Raymond air separators were typical of cement plant design when the Lyons plant was built. Many cement plants, including Lyons, have installed high efficiency separators to replace older separators for improved energy efficiency and because the cement produced has a more uniform particle size.

3.10 EMISSIONS OF OXIDES OF NITROGEN

Cement kilns process large quantities of naturally occurring raw materials in one of the world's most complicated chemical reactions. During this process, nitrogen oxides are generated and emitted from the facility. The emission rate of NOx is highly variable even absent any changes to the manufacturing operation or equipment. Normal process control parameters (i.e., those controlled by the pyroprocessing system operators) have the greatest impact on NOx emission rates.[16] Operational stability of the pyroprocessing system also has a substantial effect on NOx emission rates. Unstable operations will result in higher NOx emissions when compared to more stable operations.

Minor chemical variations in raw materials can result in different NOx emissions characteristics from cement kilns (e.g., less than a 1% change in SiO_2 in the raw materials can significantly affect cement kiln operation and therefore affect NOx emissions).

[16] In an example at Lyons, from a Lyons Plant Pyroprocess Training presentation, the section on kiln control stated that "Proper burning zone temperature is indicated by several parameters including:. . . Normal level on NOx (200 – 400 ppm)." See Lyons Plant Pyroprocess Training, CMXLYO00187057. In other words, during normal operations of the kiln system at Lyons, NOx emissions can vary by 100% when the kiln is operating within desired parameters.

There are also several different types or classes of cement products. Chemical variations between these different products and changes in kiln feed chemistry are common and can influence cement kiln operating parameters and thus NOx emissions. Short-term process transients such as kiln feed rates, fuel quality, and ambient weather can also affect NOx emissions.[17]

Cement kilns require gas phase operating temperatures that are continuously in excess of 3000°F (1650°C) in order to initiate and maintain the chemical reactions that produce cement clinker, the intermediate product. These chemical reactions require material temperatures of 2650°F to 2800°F (1450°C to 1540°C). An oxidizing atmosphere is also required in the main combustion zone of a cement kiln because of product quality considerations. If the main combustion zone temperature is too high, equipment damage will occur. If the main combustion zone temperature is too low, the cement kiln will no longer produce a quality and salable product.[18]

The following table is of uncontrolled NOx emissions from cement kilns and is taken from the EPA ACT document for NOx in Cement Manufacturing.[19] The last column was calculated by the author using the first two columns. NOx is more generally related to heat input; and, more specifically to heat input into the rotary kiln. NOx is generally not related to clinker production rates, it is related to the amount of heat required to produce the clinker.

Table 1

Kiln Type[20]	**Average (lb NOx/t clinker)**	**Average**[21] **(mmBtu/t clinker)**	**Lb NOx/mmBtu**[22]
Long Dry	8.6	4.5	1.91
Long Wet	9.7	6.0	1.62
Preheater	5.9	3.8	1.55
PH/PC	3.8	3.3	1.15

The figures in Table I above are averages. NOx emissions are highly variable on the same kiln and vary substantially between different kilns. Average values from individual kilns may be substantially different than these values. NOx emissions are very site specific and kiln specific (i.e., two kilns at the same site may have very different NOx

[17] Penta Engineering Corp., Gerald L. Young; "Report on NOx Formation and Variability in Portland Cement Kiln Systems Potential Control Techniques and Their Feasibility and Cost Effectiveness"; Portland Cement Association, R&D Serial No. 2227, Skokie, IL, 1999, p. ES-i.
[18] Penta Engineering Corp., Gerald L. Young; "Report on NOx Formation and Variability in Portland Cement Kiln Systems Potential Control Techniques and Their Feasibility and Cost Effectiveness"; Portland Cement Association, R&D Serial No. 2227, Skokie, IL, 1999, p. ES-ii.
[19] USEPA, Alternative Control Techniques Document Update – NOx Emissions from New Cement Kilns, EPA-453/R-07-006. November 2007, Table 6-1, p 36.
[20] The author has arranged the kiln types based on the degree of difficulty in operating these different kiln types, from the most difficult at the top to the easiest to operate at the bottom. This is based on the author's experience with the various kiln types.
[21] USEPA, Alternative Control Techniques Document Update – NOx Emissions from New Cement Kilns, EPA-453/R-07-006. November 2007, Table 3-3 Heat Input by Cement Kiln Type, page 13.
[22] The author has calculated the pounds of NOx per mmBtu for each of the kiln types.

emissions characteristics). For example, the author was involved in an extensive research project in 1983 to 1985 to determine what was causing NOx emissions from cement kilns. Six minute averages of more than 100 data points were collected for 270 days. Individual six minute averages on a single day ranged from 14 lb NOx/ton clinker to less than 2 lb NOx/ton clinker. Daily averages of the 240 six minute averages over the 270-day period likewise ranged from as high as 14 lb NOx/ton clinker to less than 2 lb NOx/ton clinker. This was on a long dry process kiln that had no changes to the kiln system during the period of the test. The differences were primarily because of differences in operators and in operating conditions. Below, the author has copied two graphs of NOx emissions from a Portland Cement Association document from which this information is taken.[23]

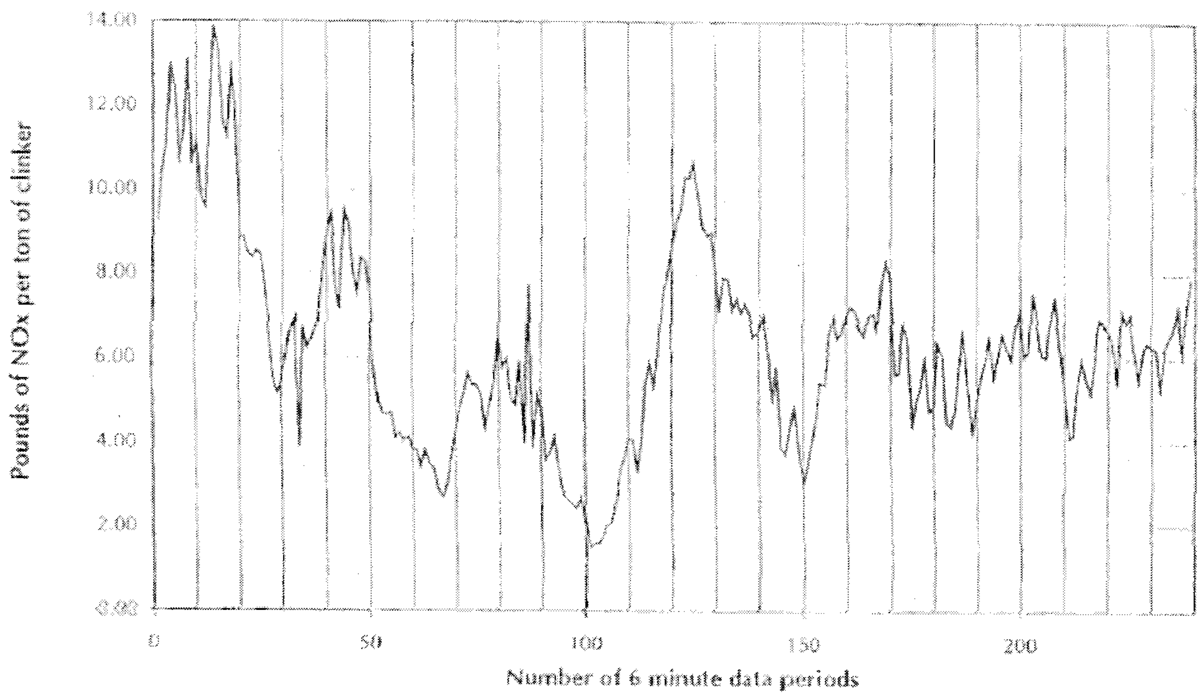

FIGURE 2 – POUNDS OF NOx PER SHORT TON OF CLINKER FOR A SINGLE DAY (240 DATA SETS – 6 MINUTE AVERAGE VALUES)

In the first graph, each of the vertical lines represents one hour. It can be seen that the NOx emissions are highly variable during the first two-thirds of the graph but are somewhat less variable in the last (rightmost) portion of the graph. This is likely the result of normal variations in operating parameters of the kiln controlled by three different operators over a 24-hour period.

[23] Penta Engineering Corp., Gerald L. Young; "Report on NOx Formation and Variability in Portland Cement Kiln Systems Potential Control Techniques and Their Feasibility and Cost Effectiveness"; Portland Cement Association, R&D Serial No. 2227, Skokie, IL, 1999, pp. 16, 17.

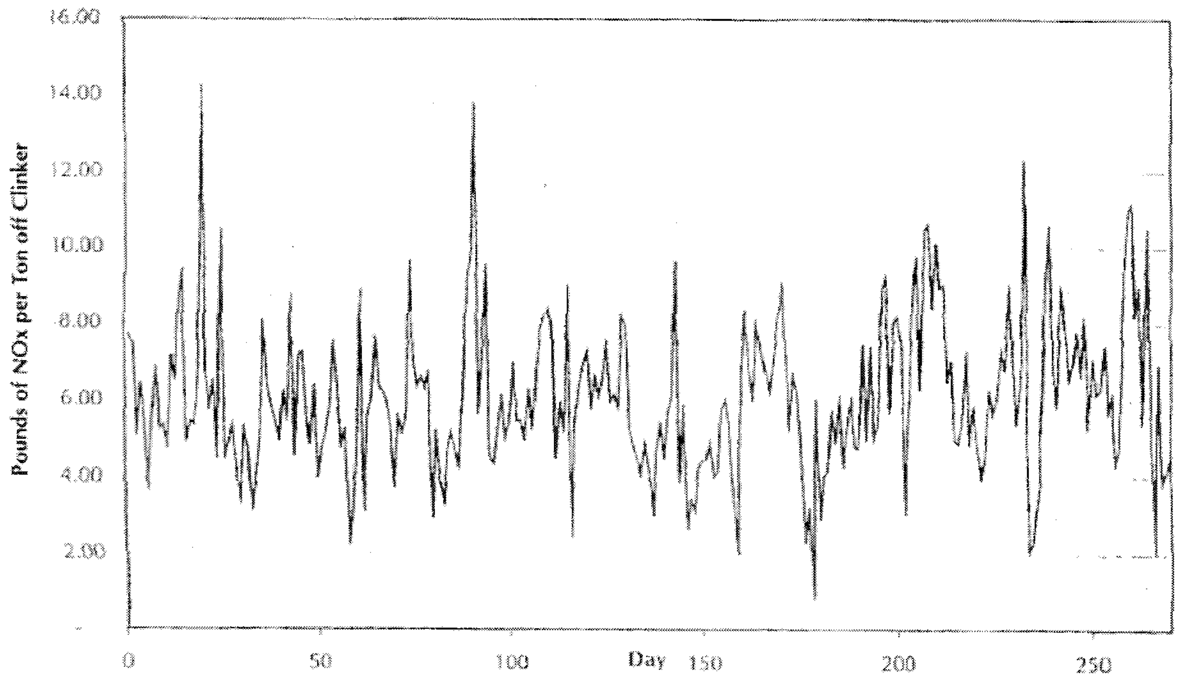

FIGURE 3 – 270 DAILY AVERAGES OF 6-MINUTE NOx (AS NO_2) PER SHORT TON OF CLINKER

The second graph displays NOx variability over a 270-day period, with no changes to the pyroprocessing system during this period. The effects of normal, long-term variations in process parameters on NOx emissions can be seen.

Despite the variability of NOx emissions displayed in the above graphs, it is possible and appropriate in certain circumstances to conduct short term testing to determine the effect of a particular change on emissions, especially on precalciner kilns which are more stable than the kiln in the tests described above.

Based on the above table (Table 1) it can be seen that for kilns that burn all of the fuel in the burning zone (i.e., long dry, long wet, and preheater kilns) the NOx is similar based on the heat input to the kiln and is not particularly based on the production rate. The difference between the three kiln types ranging from 1.9 lb NOx/ton clinker to 1.6 lb/ton clinker is explained by the differences in the difficulty in operating these types of kilns. Long dry kilns are much more difficult to operate in a stable manner when compared to preheater kilns. Preheater/precalciner (PH/PC) kilns are notable for the much lower NOx emissions relative to the other three types. This is because the fuel input is not entirely supplied to the burning zone. In calciner kilns, up to 60% of the total fuel requirements are fired in the calciner. At the Lyons plant about 50 to 55% of the fuel is burned in the calciner.[24, 25, 26]

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[24] CMXLYO00155868, Oxygen Enrichment Test, Lyons, Colorado Plant, March 1997.
[25] CMXLYO00612639, Oxygen Enrichment to Calciner Firing, Herman Tseng and John W. Lohr, CEMEX, Inc.

When NOx emissions are generally discussed units of pounds of NOx per ton of clinker are typically used. However, NOx emissions are actually more closely related to fuel input than feed input (or clinker output). For example, if a kiln system is brought up to normal operating temperature but no kiln feed is added to the process, NOx emissions will result nonetheless. NOx emissions are dependent on heat input and on the location of the heat input. If all of the heat input is in the rotary kiln, then substantial amounts of thermal NOx will be produced. If some fuel is burned in the calciner at relatively low temperatures (e.g., 1700°F versus 3400°F in the rotary kiln), then the total NOx emissions will be lower because no thermal NOx is produced in the calciner.

The Lyons kiln system will produce higher NOx emissions than a typical multistage preheater kiln with a calciner because it only has one preheater stage and consequently exhausts gases at 1500 to1600°F at the top of the preheater compared to about 700°F with a regular four or five stage preheater. However, by firing approximately 50% of the fuel in the calciner, the NOx emissions are less than if all the fuel was fired in the burning zone of the rotary kiln. The more fuel that Lyons can burn in the calciner, the lower the NOx emissions will be from the overall pyroprocessing system.

The design of the fuel firing point in the Lyons calciner tends to create a low-NOx calciner effect. Fuel is added in the riser from the rotary kiln and before tertiary combustion air is added to the calciner. This creates a reducing zone where NOx that has been formed in the rotary kiln is partially destroyed. To quote from the expert report by Walter L. Greer:[27]

> The use of riser duct firing at Lyons has the potential to lower the NOx emission factor because the kiln flue gas passes through an oxygen-deficient, fuel-rich zone in the riser duct, where free radicals react with and eliminate a portion of the NOx generated in the rotary kiln.

Indirect-fired coal systems with multi-channel burner pipes were not common use in the cement industry in 1997 to 1999. A few plants had completed some short-term experiments with Selective Non-catalytic Reduction (SNCR) but to the author's knowledge, no SNCR systems were in full-time use in the cement industry in 1997 to 1999.

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[26] CMXLYO00045539, Herman Tseng and John W. Lohr, Oxygen Enrichment, International Cement Review, May, 2001.

[27] Greer, Walter L., Expert Report of Walter L. Greer, On Behalf of the United States of America, Civil Action No. 09-cv-00019-MSK-MEH, U.S. District Court for the District of Colorado, March 1, 2011, p. 23.

4.0 CHANGES TO THE LYONS PLANT: 1997 - 2000

In nearly all cases, the cost of fuel for the pyroprocessing system is the single greatest cost for a cement plant. Accordingly, cement plants are always trying to improve energy efficiency. The Lyons plant between 1997 and 2000 made several changes to the pyroprocessing system that resulted in improvements to the energy efficiency of the system as enumerated below:

- Energy efficiency improvements to the internal geometry of the calciner
- Use of oxygen enrichment (OE) in the calciner
- Energy efficiency improvements to the clinker cooler[28]
- Installation of a booster fan on the kiln burner pipe

Due to the effects of these projects on efficiency and plant operations, the use of emission factors based on historic emissions to calculate anticipated emission increases from these projects would not be appropriate. In fact, the years following the calciner energy efficiency improvement and the use of oxygen enrichment in the calciner demonstrated increased clinker production and reduced overall NOx emissions.

4.1 PYROPROCESSING SYSTEM CHANGES: 1997

4.1.1 Efficiency Improvements to the Calciner

Following the change of the pyroprocessing system at Lyons from the original long dry kiln to a one-stage preheater, calciner kiln, a condition occurred that the plant termed "burnover." As discussed above, burnover refers to a situation where fuel fired into the calciner does not fully combust in the calciner but rather continues to combust as the gases flow into the cyclone and even at times into the conditioning tower following the cyclone. This can lead to damaged equipment and relatively high CO emissions. FCT, a reputable engineering contractor, was hired by Lyons to provide mathematical and physical modeling to improve combustion conditions in the calciner to mitigate the burnover problem.

The FCT efficiency improvements to the calciner, which were installed in the spring of 1997,[29] reduced the cross section of the calciner to increase turbulence for better air-fuel mixing. Reducing the cross section and increasing turbulence would cause the static pressure drop across the calciner to increase.[30] At a given ID fan power consumption (amp load), this would reduce the volume of air that the fan can move through the

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[28] Although not specifically discussed in this section, improvements to the clinker cooler during the 1997-2000 timeframe would have increased the fuel efficiency of the Lyons plant. The potential efficiency gains that can be realized from the types of clinker cooler improvements made at Lyons are discussed above in further detail.

[29] Case Study of Physical Modeling at Southdown's Lyons Plant, CMXLYO00402637.

[30] See Deposition of John Lohr, p.71.

system. In the case where an ID fan is "amped out" the air flow would be reduced when the static pressure of the system increases.

FCT stated that "[m]odifying the internal profile will reduce the volume of the calciner which may affect the potential production capacity of the calciner."[31] FCT also stated that CEMEX should "[c]onsult with the Fuller Company with respect to production rate implications from calciner modifications."[32] Based on these two precautionary statements, FCT was likely aware that their suggested efficiency improvement would increase the static pressure and reduce the air flow rate through the calciner.

This efficiency improvement permitted the Lyons plant to burn fuel more effectively in the calciner without the "burnover" problem (and without the corresponding high CO emissions). Because the fuel was now being burned in the appropriate place in the system, CEMEX could process more feed and produce more clinker with the same quantity of fuel. However, the total amount of fuel that could be burned in the overall pyroprocessing system would have been reduced because of the reduction in air flow due to the static pressure increase across the calciner.

It would be reasonable to conclude that this improvement to the calciner would decrease overall NOx emissions for several reasons:

- Because the total fuel that could be consumed by the pyroprocessing system was reduced, overall fuel NOx emissions would decrease.
- With less fuel being burned in the rotary kiln, thermal NOx would decrease.
- To the extent this improvement allowed fuel to be shifted from the kiln to the calciner, even further reductions in thermal NOx generated in the kiln would result.
- Improved combustion conditions in the calciner would have resulted in a higher degree of calcination, reducing the heat (fuel) requirements in the rotary kiln and thereby reducing thermal NOx formation.

In summary, while the calciner combustion chamber improvement was implemented primarily to minimize "burnover,"[33] it also improved combustion efficiency in the calciner and resulted in reduced CO and NOx emissions from the facility.

4.1.2 Induced Draft Fan Horsepower and Speed Increase

In 1997, the 1250 hp ID fan motor suffered a catastrophic failure. The 4160 volt motor shorted out electrically and caused severe damage to the motor. The pyroprocessing

[31] Fuel and Combustion Technology International, Southwestern Portland Cement Company Physical Model Study of Precalciner, 2 August 1996, number 7 in the conclusions section, p. 26, CMXLYO00453818.

[32] Fuel and Combustion Technology International, Southwestern Portland Cement Company Physical Model Study of Precalciner, 2 August 1996, p. 26, number 2.b in the recommendation section, CMXLYO00453818.

[33] See Deposition of Randal Wiley, November 10, 2010, pp. 110.

system was shut down because it cannot operate without the ID fan. This constituted an emergency repair, not a project to increase clinker production.

When the ID fan motor failed, Lyons asked Reliance (the plant's electrical supplier) to find an exact match for the motor (1250 hp, same rpm, same frame size, etc).[34] The distance from the bottom of the base of the motor to the centerline of the motor shaft had to be either the same or less than the original motor. If this distance to the centerline was greater, the motor would not have been usable without significant changes to the fan motor foundation, which would have taken many days. Reliance could not find an exact match but did find a used 1500 hp Louis Allis motor.[35] The motor was not selected because it was 1500 hp, but rather because it was the only motor available that could have been used as a replacement.

In a letter to the Emergency Breakdown Team, dated August 11, 1997, the plant manager, Mr. John Lohr, commended his team for the repairs that were completed to get a motor for the ID fan to replace the original motor that had failed. The motor failed on a Friday morning and the repairs were completed in about 80 hours, or a little more than three days. The ID fan motor replacement was clearly a plant emergency repair, so it was reasonable for CEMEX not to consider it as a PSD triggering event.[36]

The 1250 hp motor has been described as operating "amped out" at about 1030 rpm and the 1500 hp motor was eventually operated "amped out" at about 1070 rpm; however, the speed of the ID fan has also been described as being limited by vibration.[37] Vibrations in ID fans are somewhat common and are usually the result of dust buildup on the rotating fan blades. Dust will tend to build up rather uniformly but a piece of the build up will eventually break off and the fan will become out of balance resulting in vibration. If the vibration is sufficiently severe the fan speed (rpm) will have to be reduced. In extreme cases, the fan will have to be shut down and cleaned of build ups.

Assuming the ID fan had been limited to 1030 rpm and was eventually operated at 1070 rpm, then about 3.88% additional gas flow through the fan may have been realized. A 3.88% increase in the gas flow rate through the fan does not mean that 3.88% additional fuel could have been burned with a corresponding 3.88% increase in clinker production.

A significant portion of the additional flow through the fan would consist of ambient air coming through leaks[38] downstream of the calciner firing point (that do not pass through

[34] See Deposition of John Lohr, p.67.
[35] See Deposition of John Lohr, p.67, 68.
[36] See Deposition of John Lohr, p. 75 ("Again, I don't believe that I was changing a critical component. I was replacing a burned out motor with what I had available.").
[37] See Deposition of Randal Wiley, November 10, 2010, pp. 175, 176.
[38] Based on the author's experience, in a well maintained plant 10 to 20%% air leakage from the calciner firing point to the ID fan inlet is common. Because the Lyons ID fan is located rather remotely the air leakage would likely have been more than 20%. The author notes that the ID fan is located further from the preheater tower than is normally the industry practice. This is because when the preheater and calciner were installed to minimize VOC emissions resulting from the kerogens in the Lyons quarry shale, the kiln was essentially cut in half. The original baghouse was not relocated and the new ID fan was placed in the

the pyroprocessing system). It is common for such leaks to exist even on a well maintained modern kiln system. Typically in cement plants the age of the Lyons plant the ambient air leaks are much greater than 20%, but using a conservative value of 20% the air leakage would amount to 0.78% of the 3.88%. Which leaves 3.1% at 550°F that is drawn from the outlet of the calciner. Of this additional 3.1% (at 550°F) at the calciner gas outlet, a portion of it will be steam from the additional water spray required to cool the additional gases from the preheater from 1500 to 1600°F to about 550°F.

When ambient air leakage and the additional steam from the water sprays used to cool the gases from 1550°F to 500°F are considered, the net available increase in the air flow rate that could support additional production is most likely in the range of 1.5 to 2.5%, rather than the 3.88% of gases passing through the ID fan. Also, any additional fuel that could have been used would have been split approximately 50% to the calciner.

In a presentation on Riser Coal Firing, Lyons, Colorado, 04/09/02, a slide of the Kiln Throughput History attributed a daily clinker production increase of 28 t/d to the ID fan motor horsepower increase.[39] If the kiln historical production rate of 54.5 tph (as of 1/4/1998)[40] is used then the clinker production rate would be 1308 tons per day and 28 tons per day would be a 2.14% increase. This small increase would not, in the author's experience, have alerted the Lyons staff that it might significantly increase NOx emissions.[41]

4.1.3 Oxygen Enrichment, Purchased Oxygen

4.1.3.1 MARCH 1997 OXYGEN ENRICHMENT TEST

In early 1997, the Lyons plant elected to do testing of oxygen enrichment to quantify the potential increase in production and the effect on emissions, particularly on NOx emissions. The oxygen enrichment (OE) testing in 1997 was described as follows:[42]

- The participants appear to have included: Herman Tseng and Phillip Alsop from Southdown Corporate,[43] Air Products technical staff, Southdown Plant Engineers from the Knoxville and Victorville plants, an engineer from FCT, and the Lyons staff as needed.

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original ID fan location. Normally, a preheater ID fan is located at ground level, close to the preheater structure. Accordingly, the Lyons ID fan is located about 200 to 250 feet further away from the preheater tower than normal.

[39] CMXLYO00151201, Riser Coal Firing, Lyons, Colorado, 04/09/02.

[40] Memorandum, To: Dave Repasz and Bruce Ballinger, From: John Lohr, January 4, 1998, Subject: Lyons Kiln Production and Oxygen Use, CEMEX_NEIC_00008527.

[41] It is unclear from the documents the author has reviewed whether the fan vibration issues associated with the ID fan blades (which were not changed) limited the 3.88% flow rate increase at the fan that is based only on the reported rpm increase. It is possible that the vibration problems with the ID fan were exacerbated

[42] HARTWIG000006 to HARTWIG000012; USPROD01-033688, Oxygen Enrichment.

[43] The author knows both Mr. Tseng and Dr. Alsop professionally. Based on the author's knowledge of their reputation and abilities, the testing that was done in March 1997 was professionally designed and implemented. The author has reviewed the test design and it appears to be a well designed test.

- The duration was from 3/3/97 to 3/19/97 inclusive (excluding the baseline period).
- The clinker was examined by microscopic analyses to ensure that quality was maintained.
- The clinker produced while using OE was saved and ground separately to compare grindability and evaluate cement strengths.

It was the stated intention of the Lyons plant staff to maintain process conditions for the duration of the test.[44] The control room operators were instructed to alter fuel and kiln feed to respond to changes in pyroprocessing system oxygen levels.[45] Also, the Lyons plant arranged for additional staffing during the tests. The console operators (who operate the pyroprocessing system) and lab personnel were assigned to 12-hour shifts to provide double coverage, and key managers would extend and stagger their work days as necessary to insure that appropriate coverage was available for changes in test conditions or during critical phases of the test.[46]

The following data were to be collected:[47,48]

- Kiln feed rate
- Normal clinker and kiln feed testing
- Coal feed rate to the kiln
- Coal feed rate to the calciner
- Coal data (moisture, Btu, fineness)
- Secondary air temperature
- Kiln back end temperature
- Calciner exhaust temperature
- Preheater exhaust temperature
- Kiln back-end oxygen level
- Preheater exhaust oxygen level
- NOx
- Degree of calcination once every four hours
- Clinker density
- Clinker production by truck weights (eight hours per day to confirm instrumentation)

The Lyons plant found that clinker production increased about 2.5 to 4.5 tons for each ton of oxygen used, and used a figure of 3.5 tons of clinker production per ton of oxygen.[49] During the testing done by the Lyons plant in 1997, in addition to the clinker production increase the plant was also watching NOx emissions. According to available data and the

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[44] Oxygen Enrichment Test, Lyons, Colorado Plant, March 1997, CMXLYO00155869.
[45] Oxygen Enrichment Test, Lyons, Colorado Plant, March 1997, CMXLYO00155869.
[46] Oxygen Enrichment Test, Lyons, Colorado Plant, March 1997, CMXLYO00155869.
[47] Oxygen Enrichment Test, Lyons, Colorado Plant, March 1997, CMXLYO00155870-00155871.
[48] Project Outline for Oxygen Enrichment Test at Lyons, March 1997, CMXLYO00612385.
[49] Herman Tseng and John W. Lohr, Oxygen Enrichment, International Cement Review, May 2001, CMXLYO00045539.

Plant Manager John Lohr, the 1997 test showed that NOx emissions trended down[50, 51] and the effect was permanent, not transient[52] (reducing NOx).

It is my opinion that this test was well-designed and implemented, and it would have been reasonable for CEMEX to rely on the results to determine that oxygen enrichment at the Lyons facility could increase production while lowering overall NOx emissions from the pyroprocessing system.

4.1.3.2 1998 OXYGEN ENRICHMENT TEST – THE BOARDMAN TEST

In 1998, the Lyons plant, as part of its discussions with CDPHE regarding oxygen enrichment, decided to conduct another test of the effect of OE on emissions and contracted with a third party expert in statistical analyses to assist them. Although CEMEX believed the oxygen enrichment testing performed in 1997 was more representative because it lasted longer and more data were collected, because the test was conducted internally John Lohr believed that the results would not have been accepted by the State.[53] John Lohr said that he had no reason to believe that there would be a change in emissions when using OE and with the calciner efficiency improvement considered, only that the emissions might be better.[54]

A person on the CEMEX Citizen Advisory Panel was a professor at Colorado State University so John Lohr went to her and asked for a recommendation of someone that was an expert in statistical analyses and also familiar with industrial facilities.[55] This person recommended Dr. Thomas Boardman. Dr. Boardman was hired to help develop test procedures in collaboration with the Lyons plant personnel[56] and to analyze the data so that it was not prepared internally by CEMEX. That is, they wanted an impartial expert involved.[57] Developing a test protocol required a collaborative effort. The Lyons plant personnel understood their pyroprocessing system operation and Dr. Boardman was a professor of statistics and an expert in statistical analyses of industrial systems.[58] Dr. Boardman generally understood how to design a test protocol so that valid statistical analyses could be done, although he was not an expert on cement pyroprocessing systems.

A series of tests were conducted with varying system excess oxygen levels with oxygen enrichment on or off. In all, 10 different 2-hour tests were conducted in an attempt to determine the effect of OE on emissions. Selecting 2-hour tests for kiln gas analyses was

[50] See Deposition of John Lohr, p. 89.
[51] See Deposition of John Lohr, p. 95.
[52] See Deposition of John Lohr, p. 97.
[53] See Deposition of John Lohr, p. 230.
[54] See Deposition of John Lohr, p. 108.
[55] See Deposition of John Lohr, p. 228.
[56] See Deposition of John Lohr, p. 230.
[57] See Deposition of John Lohr, p. 228.
[58] Thomas J. Boardman, Department of Statistics, Colorado State University, Fort Collins, CO, CEMEX_NEIC_00008481.

appropriate because in a pyroprocessing system, kiln exhaust gas chemistry changes in response to fuel and oxygen changes rapidly (within minutes). It appears that CEMEX heated up the kiln system for this test series by instructing the console operators to target the kiln drive amps higher than normal, which is a reasonable way to stabilize the system in order to isolate the effects of the variables being tested. Also, by conducting these tests in short test series, the other operating parameters (e.g., fuel composition, temperatures, etc) could be held as steady as possible.[59]

Dr. Boardman analyzed the test data, and the results were presented to the Colorado Department of Health and Environment (CDPHE). CDPHE accepted the methodology and the test results.[60,61,62] CEMEX's letter transmitting the test results to CDPHE stated that the results confirmed their conclusion that oxygen enrichment did not increase NOx emissions.[63] In fact, CEMEX thought the test demonstrated a decrease but did not want to guarantee that decrease.[64] Dr. Boardman said that it would have been reasonable for CEMEX to believe that injecting oxygen would not increase CO, THC (total hydrocarbons), NOx, or SO_2.[65]

In my opinion, the 1998 test was a reasonable attempt to further assess the impact of oxygen enrichment on NOx emissions and provided additional support for CEMEX (and CDPHE) to conclude that oxygen enrichment did not increase NOx emissions at the Lyons facility.

Clearly, CEMEX reasonably believed at worst there would be no change in NOx and that there would possibly be a reduction in NOx emissions following the calciner efficiency improvement and the introduction of OE into the calciner. CEMEX's belief was proved valid by the NOx emissions in the years following these changes, as summarized in Table 2.

[59] During the 1998 OE test, the type of feed was changed. Based on the author's experience this would not keep the pyroprocess system as stable as possible. When feed types are changed on a pyroprocess system, it might upset the kiln; however, this does not invalidate the test series. Because each test was only two hours duration, the effects of the feed type change would have been isolated to one or possibly two of the individual tests.

[60] Letter to Ram Setharam, Air Pollution Control Division, Colorado Department of Health & Environment From John Lohr, with attachments, dated April 29, 1998, CEMEX_NEIC_00008479.

[61] See Southdown, Inc, Lyons Plant Performance Summary, 1998 Operating Year CMXLYO00038990; Letter to Dave Repasz from John Lohr, dated 1/17/98, CMXLY00045755.

[62] See Deposition of John Lohr, p. 230.

[63] Letter to Ram Setharam, Air Pollution Control Division, Colorado Department of Health & Environment From John Lohr, with attachments, dated April 29, 1998, CEMEX_NEIC_00008479.

[64] See Deposition of Stephen Goodrich, p. 148.

[65] See Deposition of Thomas Boardman, pp. 142 – 144.

Table 2

Year	Clinker[66] Tons	NOx[67] Tons	Pounds NOx/ton Clk	Year over Year Tons of NOx Reduction	Tons of NOx Reduction versus 1996
1996	430,272	1,873	8.7		
1997	458,694	1,701	7.4	172	172
1998	501,008	1,507	6.0	194	366

Using OE in the calciner was patented by Southdown (CEMEX) and is covered by two patents.[68] The use of this technology allowed the Lyons plant to increase clinker production while simultaneously reducing NOx emissions, as demonstrated in the above table. Accordingly, the Lyons plant personnel and CEMEX corporate personnel reasonably did not see any reason to pursue a PSD review because there was no increase in NOx emissions.

4.1.4 OXYGEN ENRICHMENT USING SITE-GENERATED OXYGEN

The installation of an on-site oxygen generation plant was an economic decision. The price of liquid oxygen delivered by truck was becoming uneconomical. CEMEX reviewed their options and solicited bids to purchase and install a Vacuum Swing Adsorption (VSA) plant on-site. In a report by Herman Tseng and John Lohr, they used a value of $70/ton of oxygen for purchased liquid oxygen. They awarded BOC a contract to supply a 50 ton per day VSA plant that had a total cost for oxygen generated of about $20/ton of pure oxygen equivalent (i.e., equivalent to liquid oxygen).[69] In my opinion it was reasonable for CEMEX to believe that its conclusions with respect to the effect of purchased oxygen enrichment on NOx emissions would be equally applicable to the use of plant-generated oxygen enrichment.

4.2 RAW MATERIAL DRYER TEMPERING TEE

The Lyons plant has raw materials that require pre-drying before they are proportioned into the raw mill. A natural gas rotary dryer is used to dry raw materials after they have been crushed in the primary crusher and before they are crushed in the secondary crusher.

According to Walter Greer, CEMEX installed a tempering tee to improve the performance of the rotary dryer:

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[66] United States v. CEMEX Inc. (Civ. No. 09-cv-00019-MSK-MEH, D. Colo.), CEMEX Lyons Facility Production – Historical Summary, CMXLYO 153252; CMXLYO 113390, CMXLYO 123033; CMXLYO 150080, CMXLYO 153181; CMXLYO 150088.

[67] United States v. CEMEX Inc. (Civ. No. 09-cv-00019-MSK-MEH, D. Colo.), CEMEX Lyons Facility NOx Emissions – Kiln – Historical Summary, CMXLYO 150074, CMXLYO 150081, CMXLYO 150089..

[68] United States Patent No. 6,488,765 B1, Dec., 3, 2002, Herman H. Tseng and Philip A. Alsop; United States Patent No. 6,688,883 B2, Feb. 10, 2004, Herman H. Tseng and Philip A. Alsop.

[69] Herman Tseng and John Lohr, Oxygen Enrichment to Calciner Firing, CMXLYO00612640.

> To be able to maintain a hot combustion chamber in the rotary dryer when switching between raw material components, a tempering tee was installed in the dryer baghouse ducting. This change resulted in better thermal efficiency and a higher production rate of the dryer circuit, in support of the anticipated increase in clinker production. [70]

The author visited the Lyons plant on April 14, 2011 and, as part of the plant tour, examined the dryer and ductwork to the baghouse specifically to see the tempering tee and how it was intended to function. The Lyons plant personnel pointed out a rectangular steel plate that had been welded to the ductwork above the combustion gas outlet of the dryer and told me that was where the tempering tee had been installed. Immediately above this area was a louver bleed air damper that was functioning. The Lyons plant personnel explained that they had tried the tempering tee for several years and it did not function as expected so they welded a steel plate over it years ago. Apparently, Mr. Greer mistakenly thought the louver bleed air damper was the tempering tee. The bleed air damper was installed at the time of the dryer conversion from a roaster in about 1980. The function of the bleed air damper was to protect the dryer fabric filter bags from high temperatures and avoid severe damage to the bags in the event of a temperature spike. The bleed air damper has no effect on clinker production or NOx emissions and has been part of the raw material dryer for many years. Likewise, the tempering tee would have had no effect on clinker production or NOx emissions.

4.3 RAW MILL CHANGES 1997 – 1999

Changes were made to the raw mill system between 1997 and 1999. An additional small air compressor was added to the pneumatic pump (FK pump) and the diameter of the convey line was increased from eight inches to ten inches.

During wet weather when the moisture content of the raw mill product increased, the ground raw material convey line would plug occasionally. This required plant staff to cut holes in the steel line in an attempt to clear the obstruction or the line may have had to be taken down to clean it. This can require 8 to 12 hours to accomplish and has to be done at whatever time, day or night, that it happens. Adding the third compressor and later changing the line size helped resolve this problem.

Cement plants are typically designed so that the raw mill system can produce kiln feed at a faster rate than the kiln can process. The Lyons plant reflected this design at the time the raw mill was changed. The maximum permitted kiln feed rate was 120 tph.[71] The raw mill capacity prior to this time was sufficient to accommodate the maximum rate for the kiln.[72] Moreover, even the initial planning memos for the Lyons Expansion Project

[70] Greer, Walter L., Expert Report of Walter L. Greer, On Behalf of the United States of America, Civil Action No. 09-cv-00019-MSK-MEH, U.S. District Court for the District of Colorado, March 1, 2011, p. 34.

[71] Kiln Permit, 1993, CEMEX_NEIC_00000203-207.

[72] June 1997 Monthly Report of Operations, CMXLYO00051784 (146.6tph actual raw grinding rate for the month; 124.1tph actual raw grinding rate year to date).

referred to existing raw mill production rates in the range of 122 to 130 tph.[73] The existing compressor had an even higher capacity of 130-135tph.[74][75] Thus, improving these systems would not debottleneck the kiln, because they already had more than enough capacity to supply the maximum permitted kiln feed rate.

Even if the raw mill became a bottleneck after the calciner improvements and was subsequently debottlenecked by the compressor or transfer line changes, that debottlenecking would have no effect on NOx emissions from the kiln. As discussed above, any additional kiln production enabled by the calciner improvements did not result in an increase in NOx emissions. Producing more raw feed to meet that additional kiln production capacity would also not increase NOx emissions.

Also, the raw mill system and the pyroprocessing system are not directly connected. There is a homogenizing silo and a kiln feed silo that each provides surge capacity for raw mill product storage. These two silos have a combined capacity of 5,700[76] tons, or 48 hours of kiln feed for the pyroprocessing system at the maximum permitted feed rate of 120 tph. The changes to the KF pump and convey lines were done to minimize plugging and the consequent maintenance effort required to clear the line, and would not have increased the rate of feed that could be delivered to the kiln.

4.4 ANNULUS BOOST FAN - 2000

Unstable flames in a rotary kiln adversely affect kiln operations as well as being unsafe for plant equipment and personnel. An unstable flame generally has a variable ignition point or plume length and during kiln upset conditions can result in the flame going out,[77] with an explosion resulting a few seconds later. Additional concerns with a burner pipe with a low tip velocity (e.g., less than 7000 fpm) include the possibility of the flame propagating back into the burner pipe and into the coal mill system resulting in a fire or potential explosion in the fuel preparation system[78] and unburned fuel dropping into the clinker load in the kiln which adversely affects clinker quality. The annulus boost fan was installed in 2000[79] to increase the air flow through the rotary kiln burner pipe to mitigate some of the problems inherent in low burner pipe tip velocity.

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[73] Lyons Plant Expansion, Phase I Project Scope, December 15, 1997, CEMEX_NEIC_00003300-05, at 3301; Lyons Plant Expansion, Project Scope Revised 2/5/98, CEMEX_NEIC_00003321-3324, at 3323.
[74] Lyons Plant Expansion, Phase I Project Scope, December 15, 1997, CEMEX_NEIC_00003300-05, at 3301.
[75] Lyons Plant Expansion – Phase I, Project Scope – Revised 12/12/97, CEMEX_NEIC_00003286.
[76] See CMXLYO000182189, O4 – Homogenizing / Kiln Feed System.
[77] Manias, Con G., Chapter 3.1, Kiln Burning Systems, *Innovations in Portland Cement Manufacturing*, Portland Cement Association, p. 254.
[78] The author has personal experience with a Raymond direct fired coal mill and burner substantially similar to the Raymond coal mill and burner at Lyons. What is described above happened at a plant in which the author was a Production Manager.
[79] The author was retained to examine changes at the Lyons facility as alleged in the First Amended Complaint or in the United States Experts' reports. The annulus boost fan was installed in 2000 and is technically not within the scope of the 1997-1999 time period alleged in the First Amended Complaint.

The Lyons plant stated that, "In addition, the new kiln feed metering screw in combination with the annulus boost fan appears to be dramatically improving kiln stability."[80] In the author's experience, improved flame stability will improve kiln operational stability. Kiln operational stability tends to result in lower NOx emissions, as explained below. For example, EPA has stated that:

> Process conditions that can affect NOx emissions substantially are: temperature stability, burnability of raw mix, and alkali and sulfur control. Temperature stability is important to maintain the quality of clinker and is achieved by stable flame conditions and energy efficiency.[81]

Walter Greer expands on this concept and explains that:

> Because of low velocity of the primary air and coal at the tip of the burner pipe, a direct-fired coal system such as the one in use at the Lyons Plant usually produces a long "lazy" flame in the burning zone of the rotary kiln.[82] From a process standpoint, this flame has undesirable heat transfer characteristics that can result in lower thermal efficiency for the kiln system and poor clinker quality. Impingement of a lazy flame on the material in the kiln can result in localized reducing conditions that can increase the generation of SO_2.[83]

The author agrees with Mr. Greer and notes that in the introductory section of Chapter 5 of the 1994 ACT document EPA stated that:

> For any given type of kiln, the amount of NOx formed is directly related to the amount of energy consumed in the cement-making process. Thus, measures that improve the energy efficiency of this process should reduce NOx emissions in terms of lb of NOx/ton of product.[84]

Thus, by improving the thermal efficiency and stability of the kiln system, the overall NOx emissions from the Lyons facility were likely reduced by the addition of the annulus fan while some production gains may have been realized.

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[80] CMXLYO00044075, Lyons Monthly Report of Operations, February 2001.

[81] USEPA, Alternative Control Techniques Document – NOx Emissions from Cement Manufacturing, EPA-453/R-94-004, March 1994, p. 4-10.

[82] See, Lyons Plant Expansion – Phase I Project Scope – Revised December 15, 1997 CEMEX_NEIC_00003299 AT 3304.

[83] Greer, Walter L., Expert Report of Walter L. Greer, On Behalf of the United States of America, Civil Action No. 09-cv-00019-MSK-MEH, U.S. District Court for the District of Colorado, March 1, 2011, p. 36.

[84] USEPA, Alternative Control Techniques Document – NOx Emissions from Cement Manufacturing, EPA-453/R-94-004, March 1994, p. 5-1.

4.5 FINISH MILL

The finish mill at Lyons replaced two old, obsolete air separators with a high efficiency separator. A modern high efficiency separator will typically increase mill production by about 5% on regular cements (Type I/II) and a little more on fine-grind products. In the author's experience, most plants that replace obsolete separators do so for product quality reasons. Competitors are marketing cement produced on high efficiency separators and the cement has a much more uniform particle size distribution.

In cement plants, the finish mill(s) are sized to provide sufficient grinding capacity to meet seasonal shipment patterns. Cement storage silos are also provided so that plants, such as the Lyons plant, that have highly seasonal shipping patterns can use this storage to buffer shipping demands in the peak shipping season. Also if a cement plant produces more clinker that it can grind, it is possible to sell clinker to other cement plants that have a clinker deficit. A finish mill is not a limit on kiln production because the plant can sell clinker.[85] The high efficiency separator installation was needed at Lyons because it was very important to improve product quality, especially on the fine-grind products.[86]

Like the raw mill system, the finish mill is not directly connected to the pyroprocessing system. The clinker produced is stockpiled in a covered A-frame with a 60,000 ton clinker capacity. In addition, the Lyons plant is permitted to stockpile more than 120,000 additional tons of clinker outside.[87] If the kiln produces 45,000 tons of clinker per month, this storage can hold more than three and one-half months of production without the finish mill using any clinker. The finish mill is not a bottleneck to kiln production.

Clinker storage capacities for cement plants are sized so that variations in cement shipments do not affect the pyroprocessing system. An excellent description of the use of clinker storage to separate pyroprocessing from cement grinding is provided in the Technical Review Document for Operating Permit 95OPBO082:

> From an overall perspective, the manufacturing process may be viewed as two segments – clinker production and cement production. The clinker storage allows the two processes to operate at different production rates. During periods of low demand for cement, clinker is accumulated. If cement is in high demand, the clinker production can be supplemented by purchase of clinker form other sources. The overall result is the clinker production can operate at a rather steady rate, while the cement production can operate in response to the current of projected demands.[88]

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[85] See Deposition of John Lohr, p 142. Also, the author has worked at cement plants that both sold clinker to other plants and purchased clinker from other plants.
[86] See Deposition of John Lohr, p 146.
[87] CMXLYO00607481, Colorado Department of Health, Air Pollution Control Division, Emission Permit No. 84BO369F.
[88] US00054616, Technical Review Document for Operating Permit 95OPBO082, Cathy Rhodes, September, 1999.

At Lyons, the clinker storage capacity of more than 180,000 tons effectively insulates the cement grinding segment from the pyroprocessing system. At a monthly clinker production rate of 46,000 tons, this is nearly four months of clinker production with no clinker usage by the finish mill.

In my opinion, replacing two old, obsolete air separators with one new high efficiency separator did not debottleneck the pyroprocessing system and thus would have no potential effect on NOx emissions. The installation of the high efficiency separator was primarily to improve cement product quality.[89]

[89] See Deposition of John Lohr, p. 146.

Appendix A – Qualifications

EDUCATIONAL EXPERIENCE

- BS in Chemistry, University of Missouri at Rolla, MO (now Missouri School of Science and Technology), 1970
- MS in Management, University of Redlands, Redlands, CA, 1983

PROFESSIONAL EXPERIENCE

Twenty years of experience in the Portland cement industry working as a chief chemist, production engineer, superintendent of process and production, and manager of energy and environmental affairs for major cement producers. Five years with a company that had a patented, wet sulfur dioxide scrubber that used cement kiln dust as the scrubbing reagent. Sixteen years as a process and environmental consultant to the cement industry. Author or co-author of more than 25 papers that cover cement manufacturing and emissions control from cement manufacturing facilities.

- **PENTA Engineering Corp.; St Louis, MO; 1994 – Present**
  - Provided cement companies plant audits and feasibility studies for new plants and plant expansions. Have worked closely with upper management to develop strategic plans for expansions at companies with multiple plants.
  - Assist cement companies with strategic planning for plant expansions, feasibility studies for expansion plans, and assist with preparation of presentations to upper management and Boards of Directors.
  - Develop conceptual plant designs, preliminary capital cost estimates, and assist with financial justification of expansion plans. As part of the preliminary capital cost estimates, develop, distribute, and evaluate bid documents for new cement production lines and for greenfield plant feasibility studies.
  - Audit and review of plant expansions and Greenfield cement plants to ensure proper selection of equipment and layout.
  - Process review of cement plants to increase production, improve efficiency and increase profitability.

- Financial analysis of cement plants to determine profitability, followed by implementation of special and capital projects or process/ operational changes to improve efficiency and profitability.

- Conceptual and process design of fuel firing systems and operator training on these systems.

- Responsible for the development and implementation of cement plant control room operators training programs.

- Assisted with financial justification and provided feasibility study including conceptual and process engineering for a plant modernization – conversion from wet process to 4-stage calciner and new raw mill.

- Provided feasibility study, conceptual and process engineering, site location, layout and presentations to World Bank/IFC for new 2800 mtpd clinker production line including logic for raw material proportioning and new VRM installed for cement grinding for a cement company in the Caribbean.

- Provided a feasibility study, conceptual and process engineering, and audit of possible solutions to converting a wet process kiln to semi-wet or dry process for a cement company in the Caribbean.

- Provided conceptual and process engineering for clinker production increase. Included new VRM for drying raw materials containing up to 22% moisture.

- Involved in multiple RFQ's for cement plant major process equipment. Analyzed equipment supplier bids to ensure the equipment being bid would meet the clients needs cost effectively.

- Serve as instructor for Portland Cement Association (PCA) courses "Cement Manufacturing for Process Engineers", "Kiln Process Short Course", "Cement Manufacturing Overview" and others.

- Provided conceptual and process engineering for multiple coal/coke firing systems for cement plants.

- Worked with the Portland Cement Association (PCA) and The American Portland Cement Alliance (APCA) to review hazardous air pollutants control technologies and their costs when applied to the cement industry.

- Assisted the Portland Cement Association, Canadian Portland Cement Association, and The American Portland Cement Alliance in responding to the US E.P.A. SIP call to reduce NO_x emissions. Provided technical assistance and control technology capital cost and cost effectiveness calculations.

- **Passamaquoddy Technologies; Portland, ME; 1990 – 1994**

    - Vice President, Project Development for process engineering investigations and process design as it relates to the use of an SO_2 and HCl acid gas scrubber that used cement kiln dust as the scrubbing reagent and renovates cement kiln dust for reuse in the cement industry.

- **St Mary's Peerless Cement Company; Detroit, MI; 1986 – 1990**

    - Manager of energy and environmental affairs

      Worked with the Wisconsin DNR to develop a remediation plan for a leaking underground fuel tank; with the Michigan DNR and Wayne County (MI) APCD on acquiring the necessary air permits to burn automobile tires in a rotary cement kiln as fuel; helped design a system to introduce shredded tires into a wet process kiln.

      Worked with Wayne County APCD, Detroit Fire Marshal's Office, State (MI) Fire Marshal's Office, Michigan DNR, Region V EPA, and U.S. EPA on regulations concerning the use of hazardous waste as a fuel replacement for rotary cement kilns.

    - Superintendent of Process and Production –

      Directed production related activities and production related process changes and provided substantial assistance in annual plant-wide budgeting and cost control/minimization. Was directly responsible for daily operations, scheduling, refractory repairs, and selection of refractories for a 2,000 tpd wet process kiln. Provided start-up schedules and directly supervised kiln start-ups.

- **Riverside Cement Company; Oro Grande, CA; 1978 – 1986**

    - Production Engineer – Worked on plant improvement projects for three cement plants in the southwest US. Developed projects to improve production rates, reduce power consumption, and/or reduce fuel consumption. Also, directed a research project to determine the amount of NOx reduction possible on a rotary cement kiln. Working with the South Coast Air Quality Management District (Los Angeles, CA area) and California Air Resources Board, a rule limiting NOx emissions on coal fired cement kilns in the SCAQMD was developed and implemented.

  - Production Manager, Mills – Responsible for production and maintenance on six raw mills and nine finish mills.

  - Chief Chemist – Responsible for quality control, including kiln feed chemical design, clinker chemistry, and cement quality and production scheduling. Also responsible for the plant shipping department.

- Missouri Portland Cement Co. – Sugar Creek, MO 1970 – 1978

  - Chief Chemist – Responsible for kiln feed chemical mix design, clinker chemistry, and cement quality. Also maintained raw materials, fuels and cement calculated and physical inventories.

  - Chemical Analyst and Physical Tester – Performed testing of cement for physical characteristics to ensure it met company, state and ASTM standards. Performed wet analytical chemical analyses on cement and clinker. Calibrated an X-ray machine for rapid chemical analyses.

  - Quality Control – Worked rotating shift and was responsible for routine quality control for a cement plant.

COMPENSATION IN THIS MATTER

PENTA Engineering Co., LLC is charging $250 per hour for the author's time reviewing documents, writing reports and other work as directed.

I personally receive a fixed salary that does not change based on the projects on which I work or the hours worked.

I own approximately eight percent of the stock of PENTA Engineering Corp., the parent corporation of PENTA Engineering Co., LLC.

Appendix B – Document List

References Used in the Preparation of the Report

REGULATIONS

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5 COLO. CODE REGS. §1001-5 (Reg. No. 3) (1995)

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Appendix C

PUBLICATIONS AND PRESENTATIONS IN THE LAST TEN YEARS BY GERALD L. YOUNG

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